

Networking Training Modules

Caden Kelly, John Haverlack



Module 1 Agenda: IPv4 Network Basics

- M1:
 1. **1:30pm – 1:50pm** - *Your terminal, Linux*
 2. **1:50pm – 2:20pm** - *Binary, IPv4, (ping, broadcast, subnets and their mask, etc.)*
 3. **2:20pm – 2:50pm** - *DHCP/MAC/DNS*
 4. **2:50pm – 3:00pm** - *10 minute break*
 5. **3:00pm – 3:30pm** - *NTP, LAN, Network Segmentation*
 6. **End of Module 1**

Wi-Fi Login

TrainingGroundWifi
Password: ACEP1234!?



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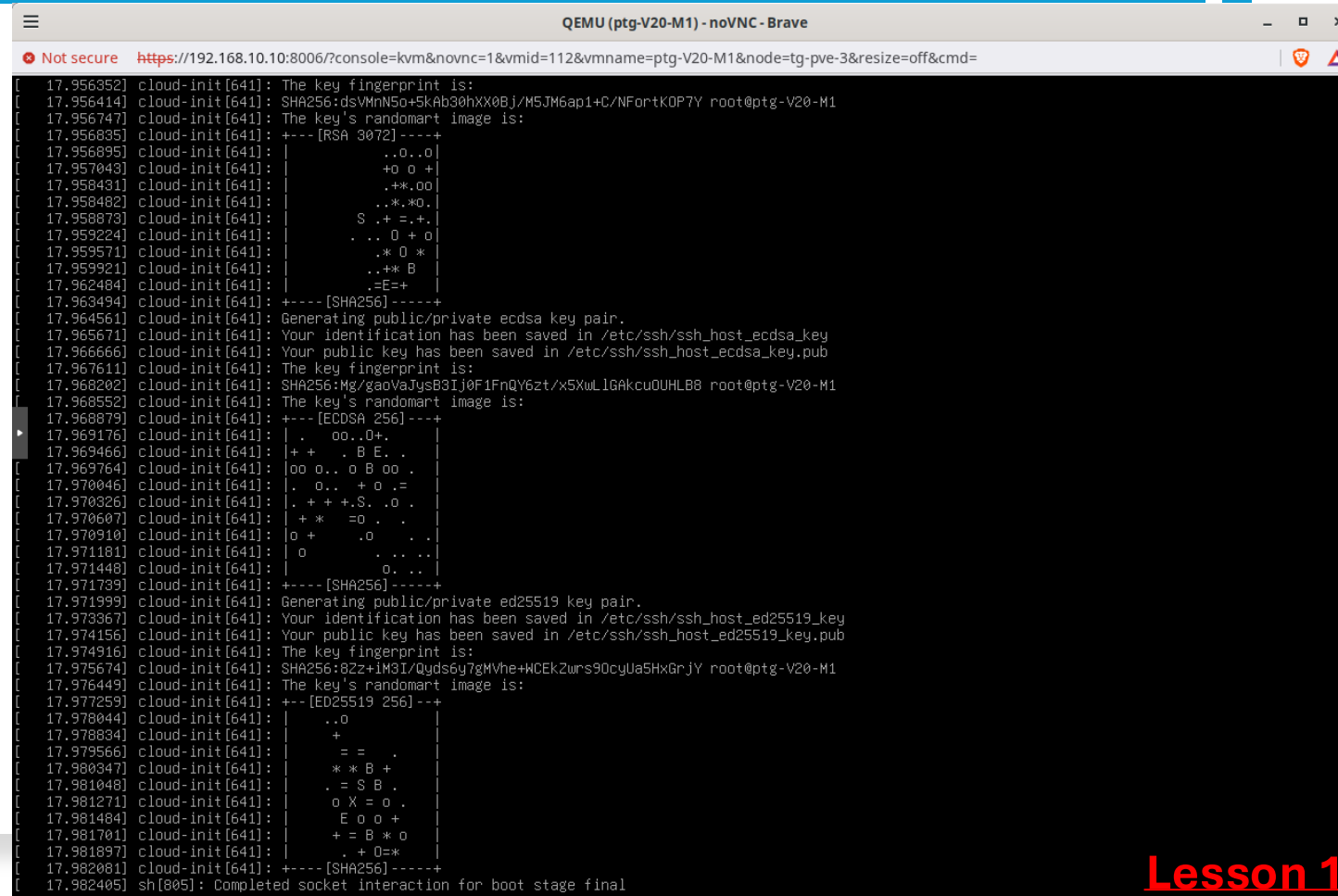
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Question: If your computer had no desktop,
no icons, and no mouse, could you still use it?

What are terminal and Linux command line?

- The terminal (AKA shell, console, etc.) is an interface that lets you interact with the computer's operating system (OS)
- File system demo
- The command line is where you can type
- Windows, MacOS, Linux, etc. are very different. Problem?



The screenshot shows a QEMU terminal window titled "QEMU (ptg-V20-M1) - noVNC - Brave". The address bar shows a URL: <https://192.168.10.10:8006/?console=kvm&novnc=1&vmid=112&vmname=ptg-V20-M1&node=tg-pve-3&resize=off&cmd=>. The terminal output shows cloud-init logs for two different key pairs (ecdsa and ed25519) being generated and saved. The logs include timestamps, the user 'cloud-init[641]', and the specific actions being performed, such as generating key pairs, saving identification, and saving public keys. The final line of the log shows 'sh[805]: Completed socket interaction for boot stage final'.

```
[ 17.956352] cloud-init[641]: The key fingerprint is:
[ 17.956414] cloud-init[641]: SHA256:dsVMN5o+5kAb30hXX0Bj/M5JM6ap1+C/NFortK0P7Y root@ptg-V20-M1
[ 17.956747] cloud-init[641]: The key's randomart image is:
[ 17.956895] cloud-init[641]: +---[RSA 3072]-----+
[ 17.956895] cloud-init[641]: | .o..o |
[ 17.957043] cloud-init[641]: | +o o + |
[ 17.958431] cloud-init[641]: | .+*.oo |
[ 17.958482] cloud-init[641]: | .+*.o. |
[ 17.958873] cloud-init[641]: | S .+ =.+. |
[ 17.959224] cloud-init[641]: | . . 0 + o |
[ 17.959571] cloud-init[641]: | .* 0 * |
[ 17.959921] cloud-init[641]: | .+* B |
[ 17.962484] cloud-init[641]: | =E=+ |
[ 17.963494] cloud-init[641]: +---[SHA256]-----+
[ 17.964561] cloud-init[641]: Generating public/private ecdsa key pair.
[ 17.965671] cloud-init[641]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 17.966666] cloud-init[641]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 17.967611] cloud-init[641]: The key fingerprint is:
[ 17.968202] cloud-init[641]: SHA256:Mg/gaoVaJysB3Ij0F1FnQY6zt/x5XwLlGakcu0UHLB8 root@ptg-V20-M1
[ 17.968552] cloud-init[641]: The key's randomart image is:
[ 17.968879] cloud-init[641]: +---[ECDSA 256]---+
[ 17.969176] cloud-init[641]: | . oo..o+ |
[ 17.969466] cloud-init[641]: | + + . B E. |
[ 17.969764] cloud-init[641]: | oo o.. o B oo |
[ 17.970046] cloud-init[641]: | . o.. + o . = |
[ 17.970326] cloud-init[641]: | . + + +S. .o |
[ 17.970607] cloud-init[641]: | + * = o . . |
[ 17.970910] cloud-init[641]: | o + .o . . |
[ 17.971181] cloud-init[641]: | o . . . . |
[ 17.971448] cloud-init[641]: | .o. . |
[ 17.971739] cloud-init[641]: +---[SHA256]-----+
[ 17.971999] cloud-init[641]: Generating public/private ed25519 key pair.
[ 17.973367] cloud-init[641]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 17.974156] cloud-init[641]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 17.974916] cloud-init[641]: The key fingerprint is:
[ 17.975674] cloud-init[641]: SHA256:8Zz+iM3I/Qyds6y7gMVhe+HCEk2urs90cyUa5HxGrjY root@ptg-V20-M1
[ 17.976449] cloud-init[641]: The key's randomart image is:
[ 17.977259] cloud-init[641]: +-[ED25519 256]--+
[ 17.978044] cloud-init[641]: | ..o |
[ 17.978834] cloud-init[641]: | + |
[ 17.979566] cloud-init[641]: | = = . |
[ 17.980347] cloud-init[641]: | * * B + |
[ 17.981048] cloud-init[641]: | . = S B . |
[ 17.981271] cloud-init[641]: | o X = o . |
[ 17.981484] cloud-init[641]: | E o o + |
[ 17.981701] cloud-init[641]: | + = B * o |
[ 17.981897] cloud-init[641]: | . + 0 = * |
[ 17.982081] cloud-init[641]: +---[SHA256]-----+
[ 17.982405] sh[805]: Completed socket interaction for boot stage final
```



192.168.10.10:8006|

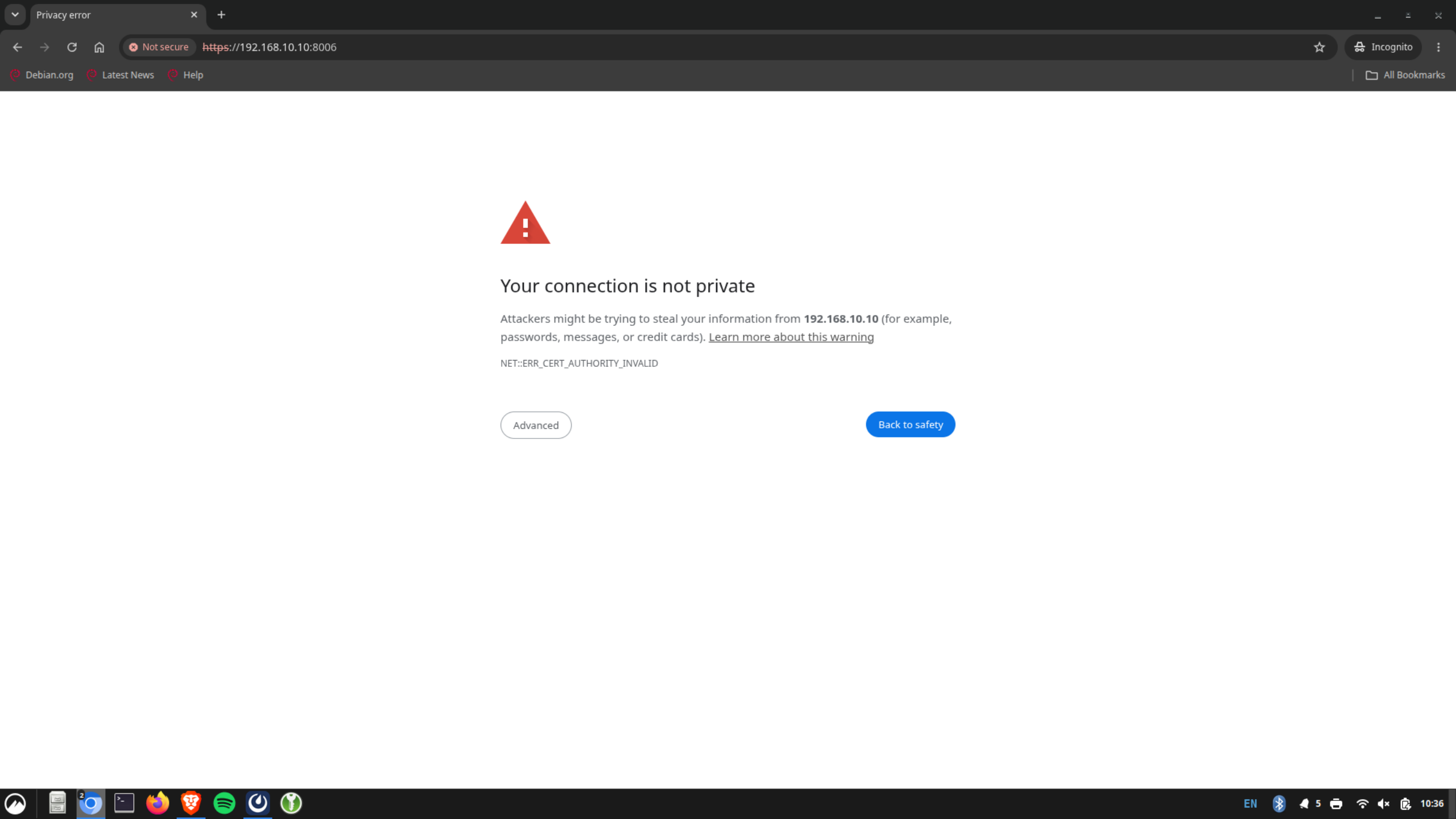
rted from Ch...

Type this into your
search bar



Ask anything, find anything...





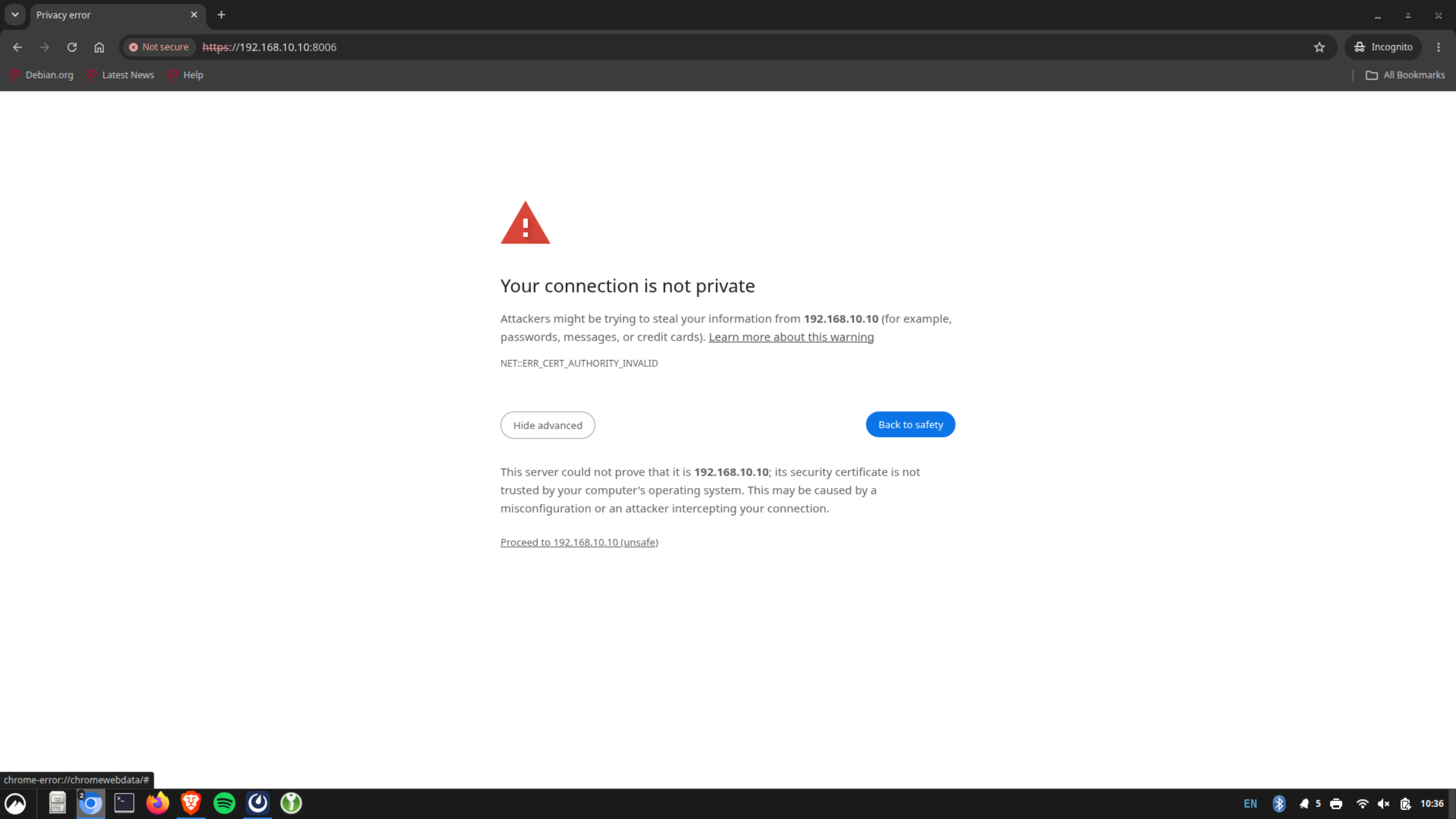
Your connection is not private

Attackers might be trying to steal your information from **192.168.10.10** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR_CERT_AUTHORITY_INVALID

Advanced

Back to safety



Your connection is not private

Attackers might be trying to steal your information from **192.168.10.10** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR_CERT_AUTHORITY_INVALID

Hide advanced

Back to safety

This server could not prove that it is **192.168.10.10**; its security certificate is not trusted by your computer's operating system. This may be caused by a misconfiguration or an attacker intercepting your connection.

[Proceed to 192.168.10.10 \(unsafe\)](#)

tg-pve-0 - Proxmox Virtual En

192.168.10.10:8006/#v1:0:18:4:.....

DocumentationCreate VMCreate CT

Server ViewDatacenter

Proxmox VE Login

User name: student1

Password: student1

Realm: Linux PAM standard authentication

Language: English - English

Save User name: Login

TasksCluster log

Start Time ↓	End Time	Node	User name	Description	Status
--------------	----------	------	-----------	-------------	--------

ENBluetoothWi-FiPower11:15

Everyone's username and password will be handed out and be student# where # is assigned to you

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Lesson 1

Server View

▼ Datacenter

tg-pve-0

tg-pve-1

tg-pve-2

tg-pve-3

Node 'tg-pve-0'

Search

Task History

Subscription

Reboot

Shutdown

Shell

Bulk Actions

Help

Search:

Type ↑	Description	Disk usage...	Memory us...	CPU usage	Uptime	Host CPU ...	Host Mem...	Tags
qemu	101 (ptg-V20-Static)				-			m1
qemu	102 (ptg-V20-M1)				-			m1
qemu	105 (ptg-V20-M1)				-			m1
qemu	120 (ptg-swm2-20)				-			m2
qemu	121 (ptg-swm2-21)				-			m2
qemu	122 (ptg-swm2-22)				-			m2
qemu	600 (V20Student1-OPNsense)				-			m2
qemu	100 (ptg-V20-M1)				-			

tg-pve-0 if you are student 1,

tg-pve-1 if student2

tg-pve-2 if student3

Tge-pve-3 if student4

tg-pve-0 - Proxmox Virtual En

192.168.10.10:8006/#v1:0:=node%2Ftg-pve-0:4:.....

☆

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📄

👤

☰

PROXMOX

Virtual Environment 9.2.3

Search

Documentation

Create VM

Create CT

student1@pam

Server View

▼

⚙️

▼

Datacenter

▼

tg-pve-0

101 (ptg-V20-Static)

102 (ptg-V20-M1)

105 (ptg-V20-M1)

120 (ptg-swm2-20)

121 (ptg-swm2-21)

122 (ptg-swm2-22)

600 (V20Student1-OPNsense)

100 (ptg-V20-M1)

tg-pve-1

tg-pve-2

tg-pve-3

Node 'tg-pve-0'

Search

Task History

Subscription

Reboot

Shutdown

Shell

Bulk Actions

Help

Search:

Type ↑	Description	Disk usage...	Memory us...	CPU usage	Uptime	Host CPU ...	Host Mem...	Tags
qemu	101 (ptg-V20-Static)				-			m1
qemu	102 (ptg-V20-M1)				-			m1
qemu	105 (ptg-V20-M1)				-			m1
qemu	120 (ptg-swm2-20)				-			m2
qemu	121 (ptg-swm2-21)				-			m2
qemu	122 (ptg-swm2-22)				-			m2
qemu	600 (V20Student1-OPNsense)				-			m2
qemu	100 (ptg-V20-M1)				-			

Tasks Cluster log

Start Time ↓	End Time	Node	User name	Description	Status
--------------	----------	------	-----------	-------------	--------

Student	VM ID
Tendai	101
Kyle	102
Michael	105

We will pass out sheets with your username, password, and VM for this part.

PROXMOX Virtual Environment 9.2.3

Server View

Virtual Machine 109 (ptg-V20-M1) on node 'tg-pve-2' **m1**

Start Shutdown Migrate Console More Help

Hour Maximum Average

109 (ptg-V20-M1) **m1**

Summary

Console Hardware Cloud-Init Options Task History Snapshots Firewall

ptg-V20-M1

Status stopped

HA State none

Node tg-pve-2

CPU usage 0.00% of 2 CPUs

Memory usage 0.00% (0 B of 512.00 MiB)

Host memory usage 0 B

Bootdisk size 50.00 GiB

IPs Guest Agent not running

Notes

deb12-tmpl-2026-06-10

This Template added Cloud Init to the system such that the hostname is automatically configured for newly cloned VM's.

Cloud Init also add the acepadmin user for John's access.

CPU Usage

Network Traffic

2

3

1

1. Press "start" in the top left corner, and once your VM has a green checkmark next to its name, press "console".

Out Your Linux Cheat Sheets

Command	What It Does
<code>pwd</code>	Print working directory that shows where you are
<code>cd</code>	Change directory (<code>cd /path</code> = absolute, <code>cd path</code> = relative, no leading slash)
<code>ls</code>	List files in the current directory
<code>find</code>	Search for files by name, type, or location
<code>grep</code>	Search text inside files for a pattern
<code>\ </code> (pipe)	Sends output of one command into the next command
<code>cat</code>	Print entire file contents to the screen
<code>nano</code>	Text editor for creating/editing files
<code>touch</code>	Create an empty file
<code>sudo</code>	Run a command with admin/root privileges
<code>ip a</code>	Show network interfaces and IP addresses
<code>ping</code>	Test connectivity to another host
<code>ip route</code>	Show the routing table
<code>traceroute</code>	Show the path (hops) packets take to reach a destination
<code>nslookup</code>	Look up the IP address behind a domain name (DNS lookup)
<code>ssh user@host</code>	Log into a remote computer
<code>scp file user@host:path</code>	Copy a file from your machine to theirs



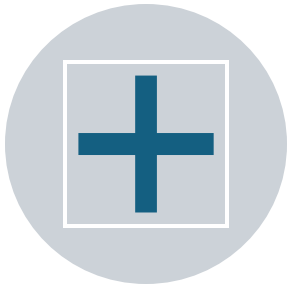
Linux Command Line Practice

Demonstration in terminal. Please follow along!

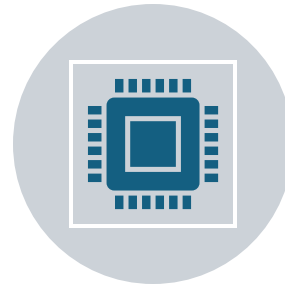


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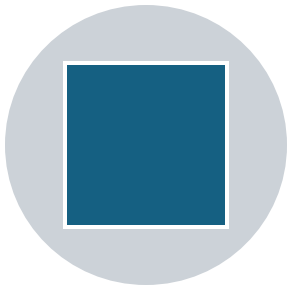
What is binary?



Binary and decimal are two ways of counting.



Computers store and transmit their info in binary (base-2)



This is how your online internet address is configured (IP Address)



Computers read your IP address, packet information, etc. in binary

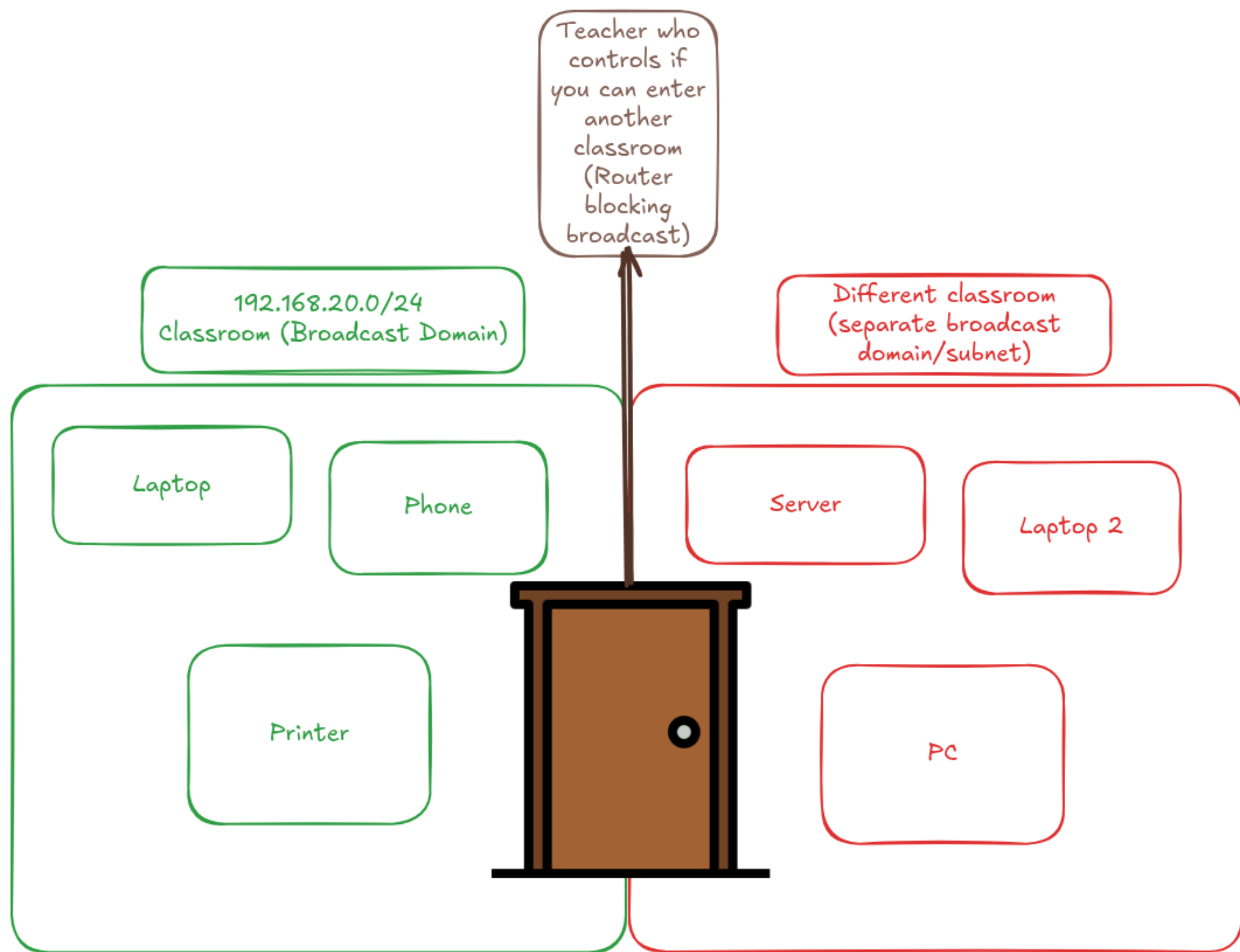
IPv4 (Internet Protocol)

How the internet recognizes your computer



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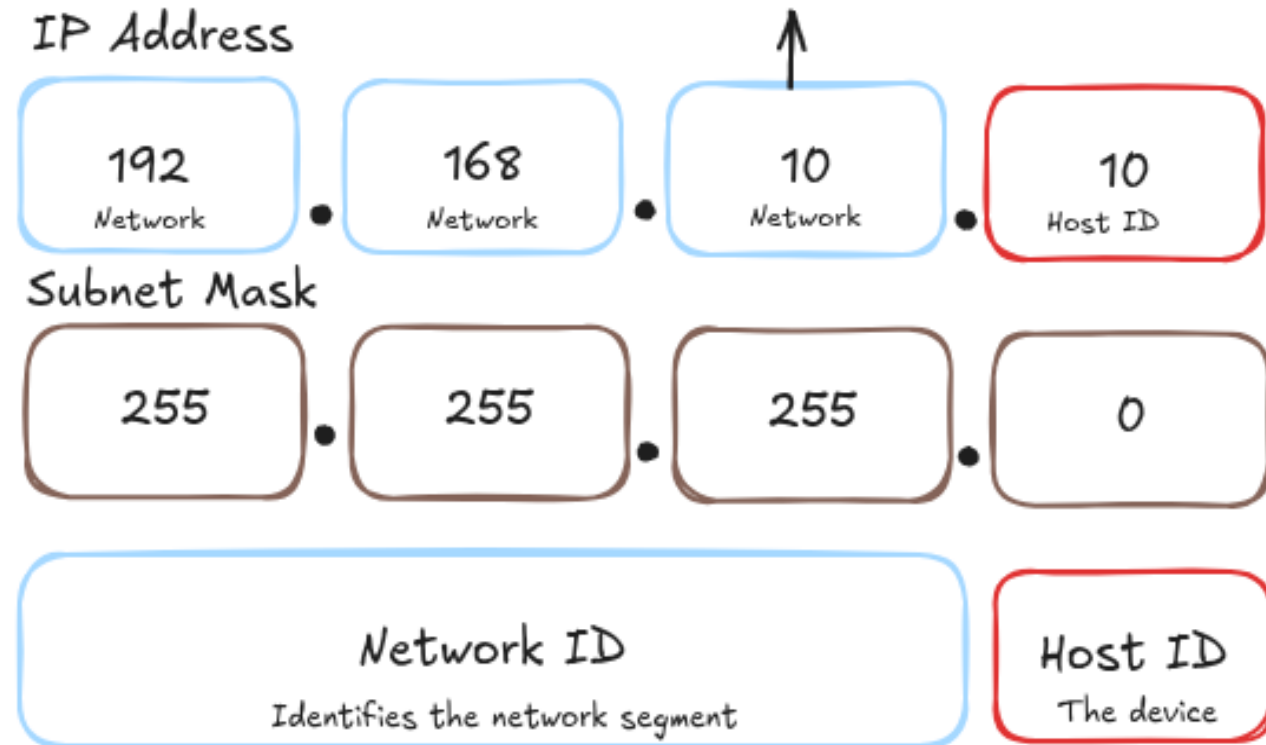
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Example IP

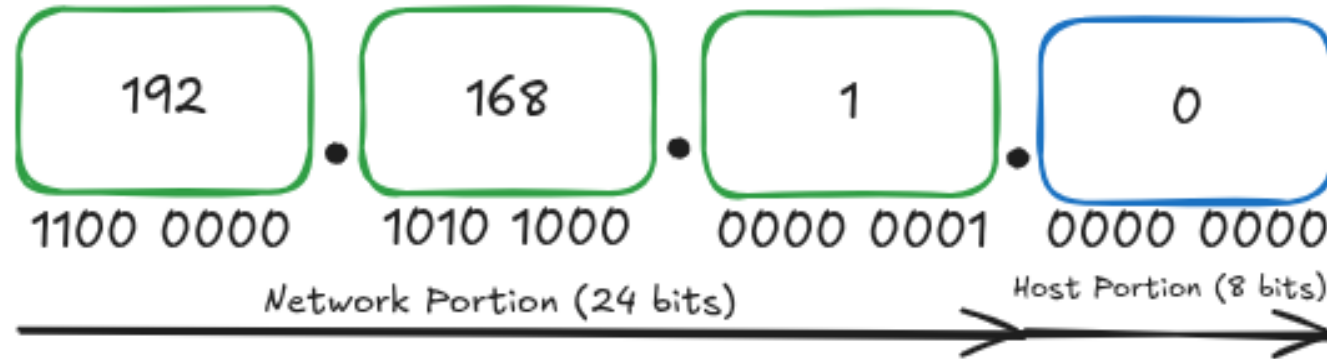
- Subnets are like rooms!!!
- 8 bits per section
- This is a 24 bit subnet mask; 8 bits left for devices on this subnet

Often change the end of the network id to make different "subnets" & refer to them by the part we change, such as the .10 subnet, but really the entire Network ID is a subnet



CIDR Notation Explained: 192.168.1.0/24

Network bits vs Host bits



Subnet Mask

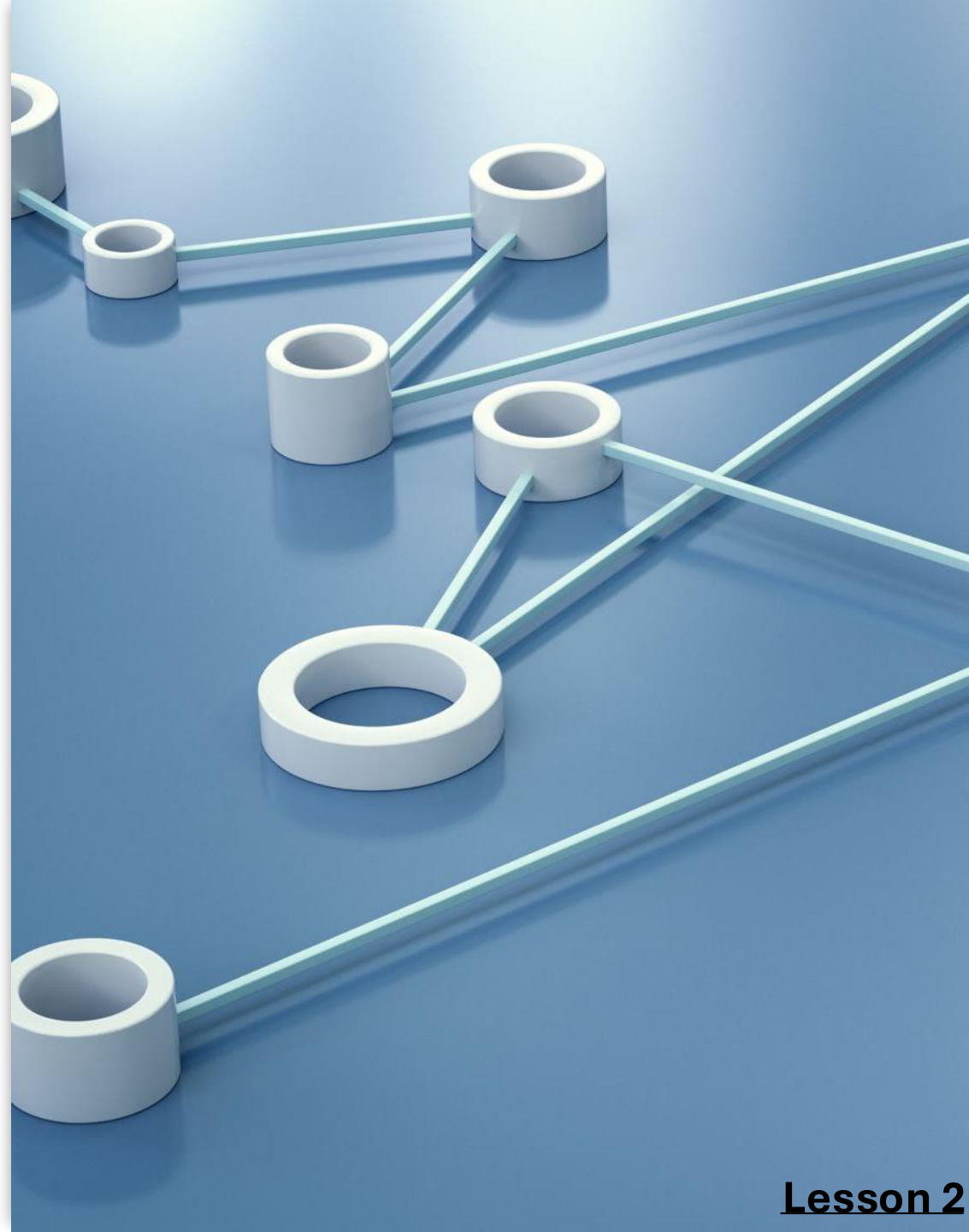


*The /24 means the first 24 bits are used for the network portion leaving 8 bits for host addresses ($2^8 = 256$ total host IDs)



Information Transfer

- When you send online information, it goes to a subnet
- Routers keep tables of where to send info based on network ID, not host addresses
- What issues arise if routers used host ID's instead?



Internal vs External IPs

Internal



- Used inside your own network
- Not routable on public internet

External
(public)

- Globally unique
- Routable across the public internet
- Assigned by ISPs

- **Network address translation (NAT):** your router swaps addresses at the edge, so every internal device shares the router's IP, so their own IP and MAC are hidden



Broadcast Domain

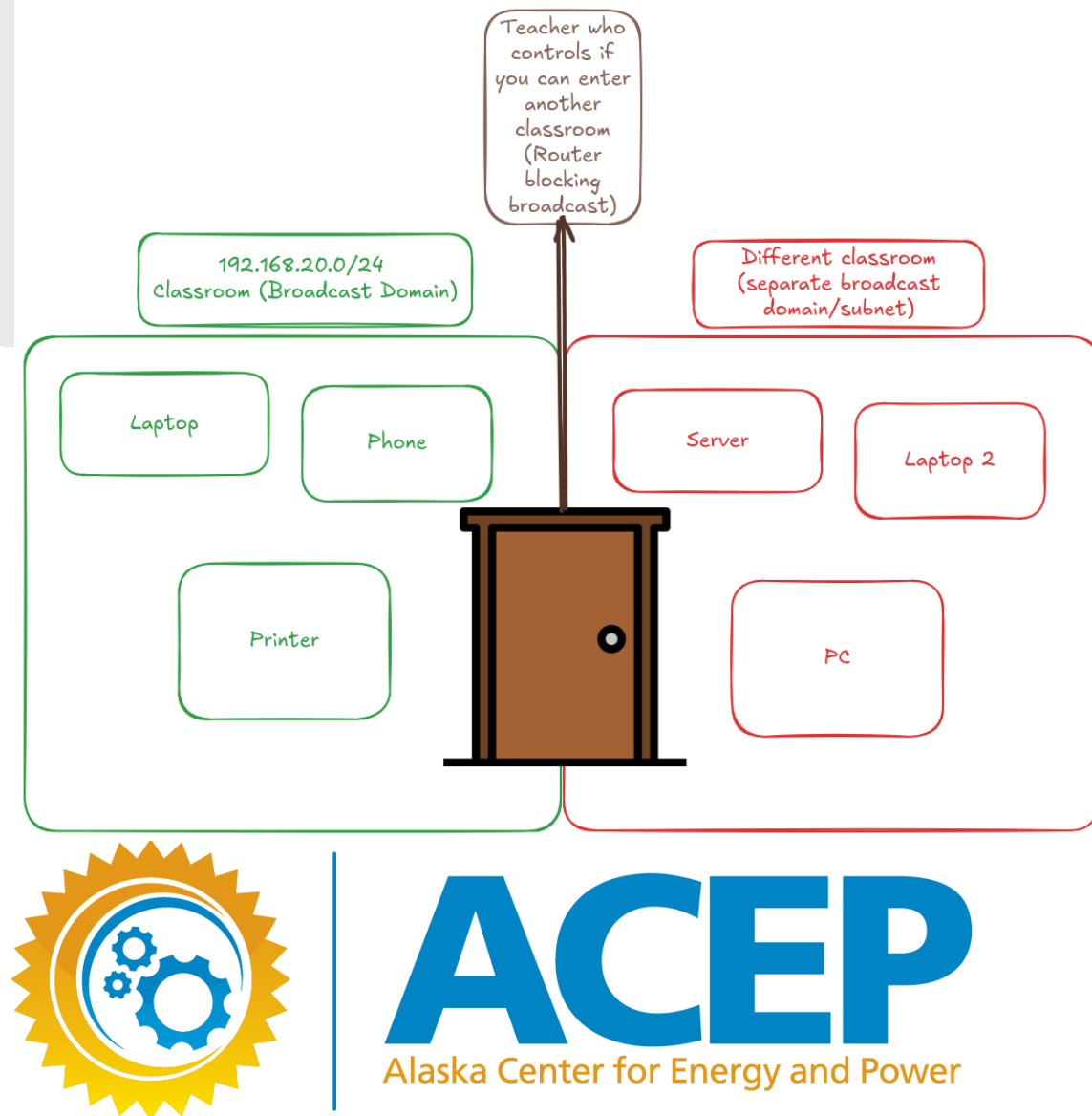
A broadcast is a message one device sends to everyone on the same network subnet

Room analogy and broadcast domain, ex. ip a

All devices in our example of the 192.168.20.xxx are all in the same room

Routers doors/walls

Smaller rooms stop devices from hearing yells that aren't meant for them





Gateway IPs

- Connects your LAN to everything else
- Inbound/Outbound traffic goes through the gateway first
- No gateway = local traffic still works, but nothing further
- Usually on a .1 subnet; 192.168.20.1



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Setting our own IPs statically

```
student1@ptg-v20-M1:~$ ip a
```

sudo nano /etc/network/interfaces

```
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug ens18
iface ens18 inet dhcp
```

Student Name	Tendai	Kyle	Michael
IP Address	192.168.20.30	192.168.20.31	192.168.20.32

```
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug ens18
iface ens18 inet dhcp
```



```
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug ens18
iface ens18 inet static
    address 192.168.20.54
    netmask 255.255.255.0
    gateway 192.168.20.1
    dns-nameservers 192.168.20.1
```

Raise your hand once it is typed I will check your file

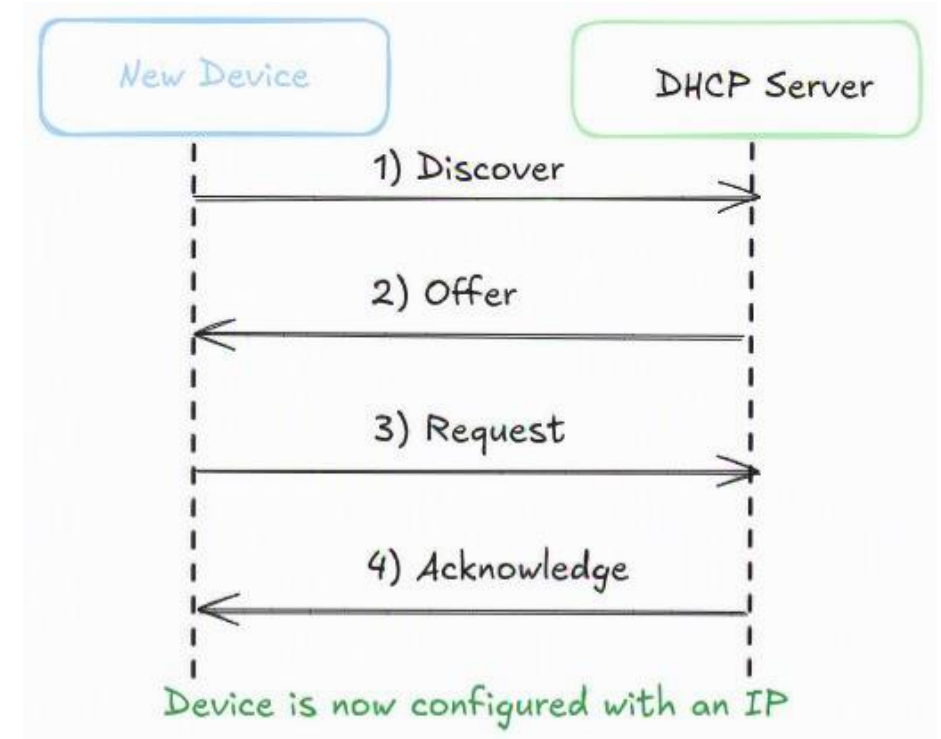
To exit and save, type: "**Ctrl+X**" -> "**y**" -> "**enter**" button

Type: **sudo systemctl restart networking**

Type **ip a**

DHCP (Dynamic Host Configuration Protocol)

- Restaurant waiter giving you a table
- DHCP is how a device automatically gets an IP address
- Without the address, information doesn't know where to go
- Range, Lease
- DHCP Address collision



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DHCP Ranges Visual - GLiNet



Why may we want a DHCP address instead of statically assigning each device?



Rollback, Re-login

If DHCP randomly assigns an IP from within a range, why is it the same as before?



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Explanation...

Library book return policy

DHCP leases vs new per connection

Within lease time, DHCP **tries to give it the same IP** if available

How does this stabilize the network?

Unavailable IP = DHCP assigns another within its range

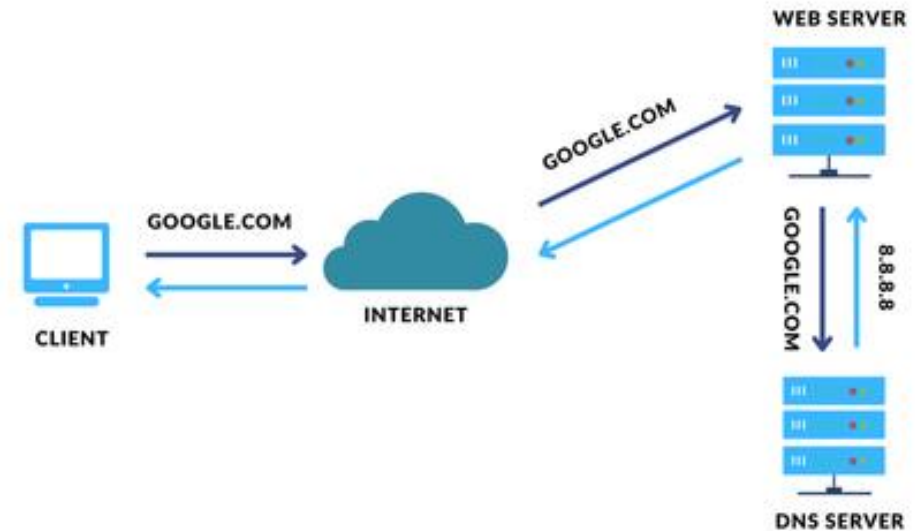
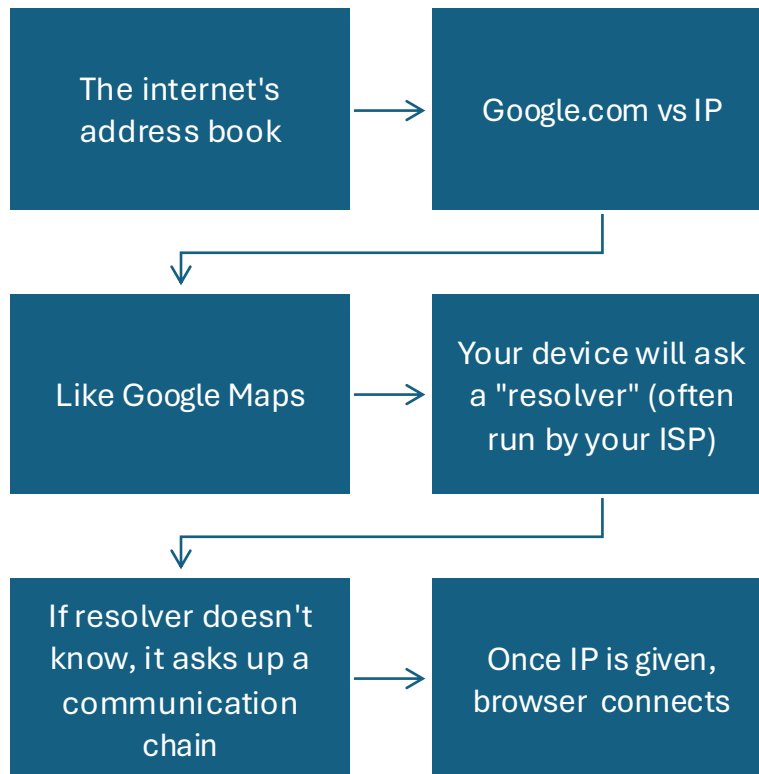


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Checking our MAC addresses

DNS (Domain Name System)



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We can see the translation and the route being taken in real time

- Type "host google.com" and "nslookup google.com"
- What are they doing?

```
student1@ptg-V20-M1:~$ host google.com
google.com has address 142.250.73.110
google.com mail is handled by 10 smtp.google.com.
google.com has HTTP service bindings 1 . alpn="h2,h3"
student1@ptg-V20-M1:~$ nslookup google.com
Server:          192.168.10.1
Address:         192.168.10.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.73.110
```



We can see the translation and the route being taken in real time

- Type "host google.com" and "nslookup google.com"
- What is it doing?

```
student1@ptg-V20-M1:~$ host acep.uaf.edu
acep.uaf.edu has address 137.229.114.150
student1@ptg-V20-M1:~$ nslookup acep.uaf.edu
Server:      192.168.10.1
Address:     192.168.10.1#53

Non-authoritative answer:
Name:   acep.uaf.edu
Address: 137.229.114.150
```

Try both these commands with **"acep.uaf.edu"** instead of google.com. What do you notice?

DNS lookup being resolved



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We can see the translation
and the route being taken
in real time

- Type "traceroute google.com"

```
student1@ptg-V20-M1:~$ traceroute google.com
traceroute to google.com (142.250.73.110), 30 hops max, 60 byte packets
 1  192.168.20.9 (192.168.20.9)  0.393 ms  0.371 ms  0.360 ms
 2  console.gl-inet.com (192.168.10.1)  1.607 ms  1.651 ms  1.563 ms
 3  192.168.9.1 (192.168.9.1)  3.169 ms  3.154 ms  3.133 ms
 4  10.25.156.1 (10.25.156.1)  5.506 ms  4.047 ms  4.781 ms
 5  10.5.222.17 (10.5.222.17)  3.422 ms  3.405 ms  3.380 ms
 6  10.5.222.13 (10.5.222.13)  3.356 ms  2.161 ms  2.127 ms
 7  10.5.222.3 (10.5.222.3)  2.098 ms  3.052 ms  2.243 ms
 8  10.5.222.2 (10.5.222.2)  2.227 ms  2.680 ms  2.646 ms
 9  172.16.0.227 (172.16.0.227)  3.222 ms  3.173 ms  3.155 ms
10  137.229.252.161 (137.229.252.161)  3.943 ms  3.917 ms  4.074 ms
11  137.229.255.232 (137.229.255.232)  43.689 ms  43.661 ms  43.638 ms
12  ae17--202.icar-sttl1-1.infra.pnw-gigapop.net (209.124.190.216)  43.616 ms  43.588 ms  43.650 ms
13  lo0--200.icar-sttl1-4.infra.pnw-gigapop.net (209.124.190.11)  43.615 ms  43.589 ms  43.566 ms
14  74.125.51.244 (74.125.51.244)  43.544 ms  43.519 ms  43.498 ms
15  * * *
16  142.250.233.134 (142.250.233.134)  43.747 ms  142.251.50.242 (142.251.50.242)  45.157 ms  142.251.55.202 (142.251.55.202)  43.851 ms
17  pnseaa-an-in-f14.1e100.net (142.250.73.110)  43.830 ms  142.251.64.87 (142.251.64.87)  43.813 ms  pnseaa-an-in-f14.1e100.net (142.250.73.110)  43.799 ms
student1@ptg-V20-M1:~$ _
```

We can see the translation
and the route being taken
in real time

- Type "tracert acep.uaf.edu"

```
1 192.168.20.9 (192.168.20.9) 0.399 ms 0.418 ms 0.403 ms
2 console.gl-inet.com (192.168.10.1) 1.516 ms 1.493 ms 1.595 ms
3 192.168.9.1 (192.168.9.1) 2.768 ms 2.738 ms 2.713 ms
4 10.25.156.1 (10.25.156.1) 4.835 ms 5.450 ms 4.798 ms
5 10.5.222.17 (10.5.222.17) 3.298 ms 3.276 ms 3.254 ms
6 10.5.222.13 (10.5.222.13) 3.236 ms 2.649 ms 2.616 ms
7 10.5.222.3 (10.5.222.3) 2.600 ms 2.364 ms 2.334 ms
8 10.5.222.2 (10.5.222.2) 2.309 ms 2.608 ms 2.572 ms
9 172.16.0.227 (172.16.0.227) 3.250 ms 3.229 ms 3.205 ms
10 137.229.114.2 (137.229.114.2) 3.903 ms 3.882 ms 3.857 ms
11 137.229.114.150 (137.229.114.150) 3.836 ms 3.815 ms 3.794 ms
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
```



10 Minute Break



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NTP (Network Time Protocol)

- NTP is how devices agree what time it is.
- Why might this be important?
- Time comes from a small number precise reference clocks
- Your device asks a local timeserver, then matches its clock



Testing!

TYPE IN
"DATE"

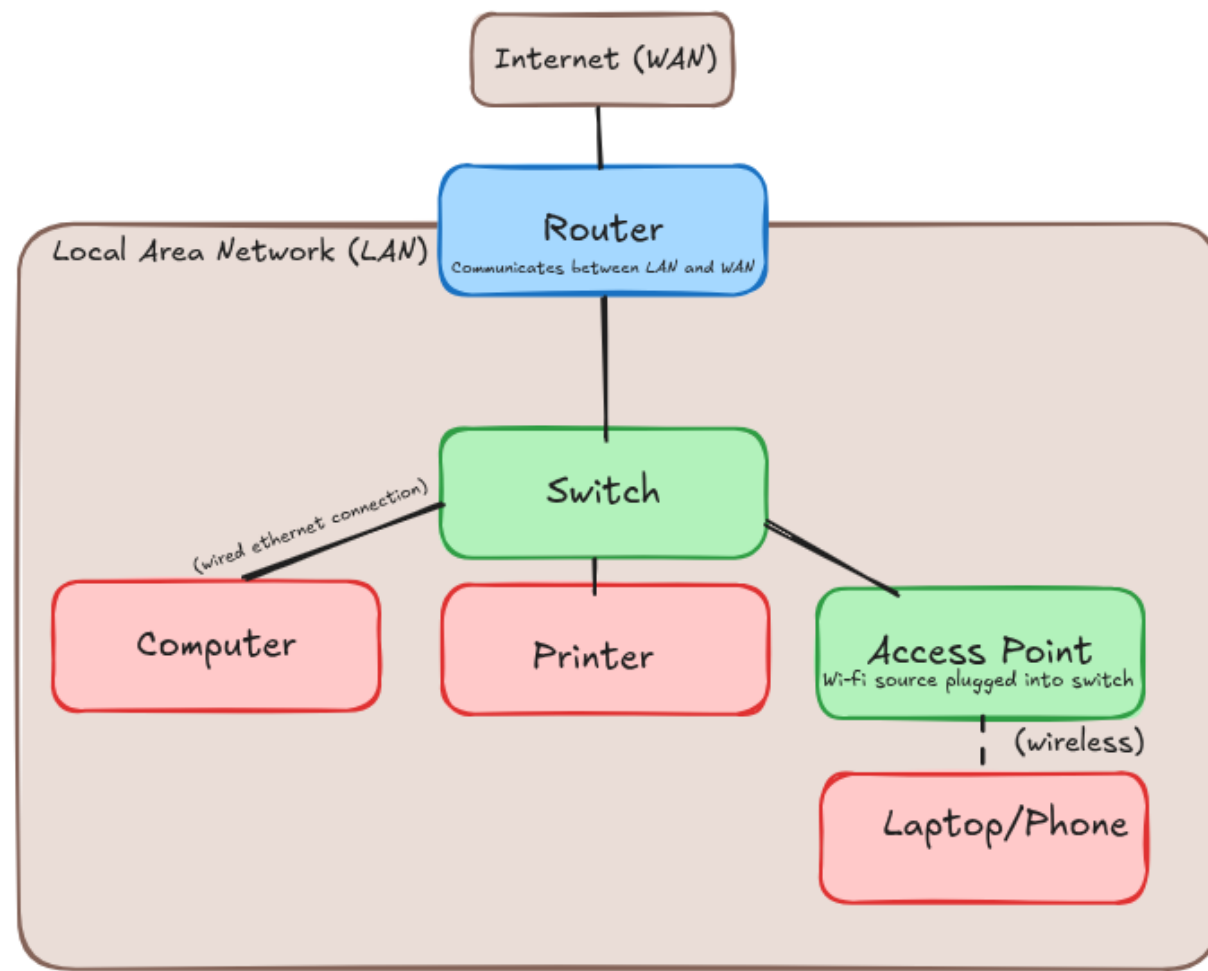
WHAT DO
YOU NOTICE?



LAN (Local Area Network)

- Classroom; what is each part
- A LAN is a network confined to one physical space
- Devices have wired or wireless connections
- Router handles edge communication
- WAN vs LAN; multiple LANs

Note: A VLAN is a virtual LAN defined to keep traffic separate without need for extra hardware. I did this for your training ground through managed switches, which work on layer 3 of the OSI



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Handouts!

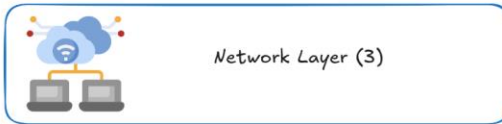
Lesson 4

Layers 4-7 Redacted

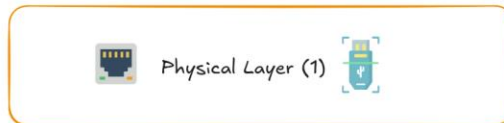
•
•
•

Handles data routing, forwarding, and addressing. Determines optimal path for data to reach destination. Uses protocols like UDP, TCP.

 **ACEP**
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Portable Training Ground
OSI Layers 1-3 Diagram
2026-07-21
Caden Kelly



Facilitates reliable data transfer across physical network links. Provides error detection and correction. Manages how data is placed onto the network medium. Protocols include ethernet and PPP.



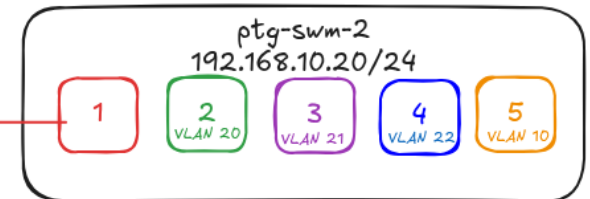
Transmit raw bitstreams over the physical medium. Handles electrical signals, cable types, and data rates.



Portable Training Ground
Student Station Diagram
2026-07-21
Caden Kelly

- The VLAN IPv4 will be on this subnet:
-> 192.168.XX.ZZZ Where XX is your VLAN and ZZZ is the DHCP address given
- Each VLAN is one student in the WSSH

Physically Cabled to Trunk switch that carries all VLANs



Physical Training Ground "Tour"

Come up to the training
ground

We will only be focusing on
OSI layers 1-3

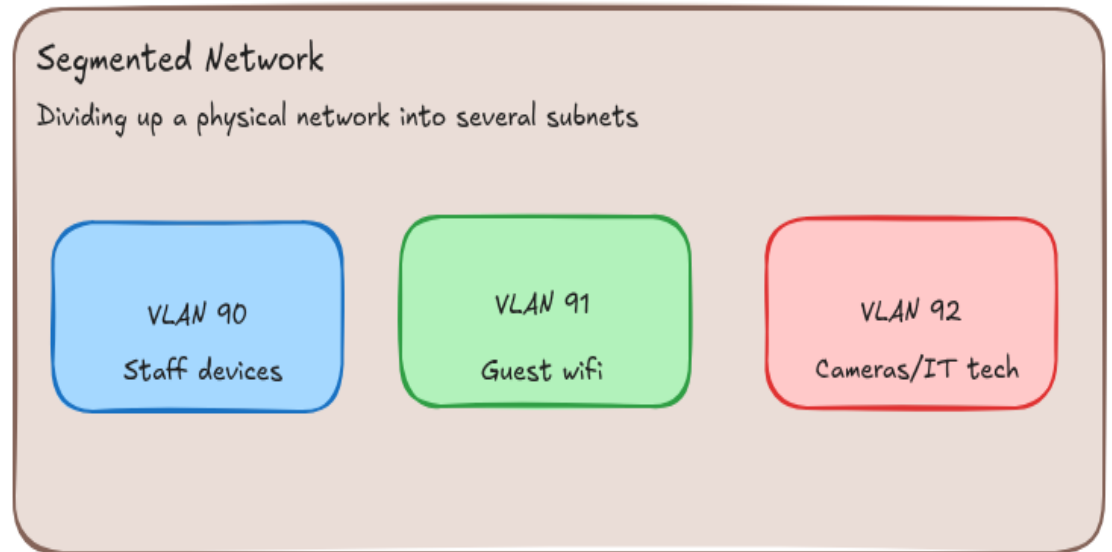
**Managed vs unmanaged
switches**

Why do we have so many
different LANs



Network Segmentation

- Store: cameras vs guests vs employees
- Splitting a network into smaller networks
- Own subnet and broadcast domain
- Used to keep different kinds of traffic apart.
- A router decides what traffic is allowed to cross



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Why is this a good idea?

Discuss with a partner then raise hands to answer



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Benefits

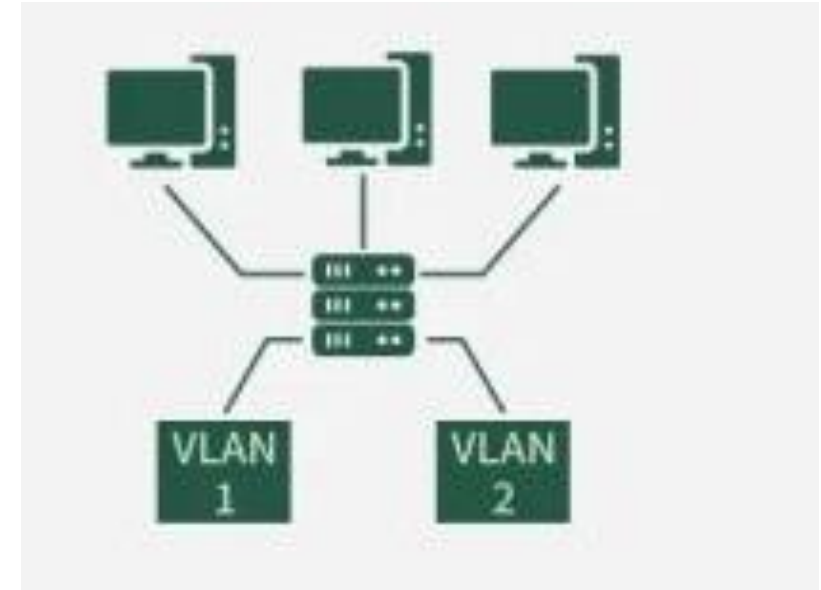
Better security

Better performance; less traffic to wade through

Easier troubleshooting

More organized

A focus of our second module - firewalls



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Activity Pt. 1

**Find someone
in your group
and try to ping
their IP**

"ping 192.168.xx.xxx" fill in the x's with their ip information
What do you notice



**Get a group
member's IP
and "tracert
<insert their IP>**

What do you notice?





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Logging Into Different Subnets WSSH

What are SSH (Secure Shell), WSSH?

- A service your computer has. Services run on virtual "ports"
- Post office/box analogy
- Secure remote access tool requiring a login
- If allowed you can log into the root (administrative) terminal of another computer
- WSSH is a public software allowing you to do this through a web browser (Web-SSH)



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```
File Edit View Search Terminal Help
```

```
cckelly2@acep-infosec-2:~$ ssh root@192.168.10.1
The authenticity of host '192.168.10.1 (192.168.10.1)' can't be established.
ED25519 key fingerprint is SHA256:PeXOEiluMX/NADAy5V7LMrAI/eusZRgnqpxlItUqotU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.1' (ED25519) to the list of known hosts.
root@192.168.10.1's password:
```

```
BusyBox v1.36.1 (2026-04-13 07:09:34 UTC) built-in shell (ash)
```

```
- . - - - - - | - - - - - | - - - - - | - - - - - |
|_| W I R E L E S S   F R E E D O M
```

```
-----
ApNos-9e9cd9a7-devel
OpenWrt 23.05-SNAPSHOT,
-----

root@GL-AXT1800:~#
```

WSSH Login

<http://192.168.10.219:8888/>



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Login Directions

Everyone will have the same username/password as the sheet we gave out earlier

The table shows your hostname to put in next to your name

Student	Hostname
Tendai	192.168.20.118
Will	191.168.21.160
Michael	192.168.22.104
Filler	192.168.30.177
Filler	192.168.31.110
Filler	192.168.32.125



Activity Pt. 2

Find
someone in
your group
and try to
ping their
new IP

"ping 192.168.xx.xxx"

What happened?

Get a group
member's
IP and
"tracert
<insert
their IP>

What do you notice?



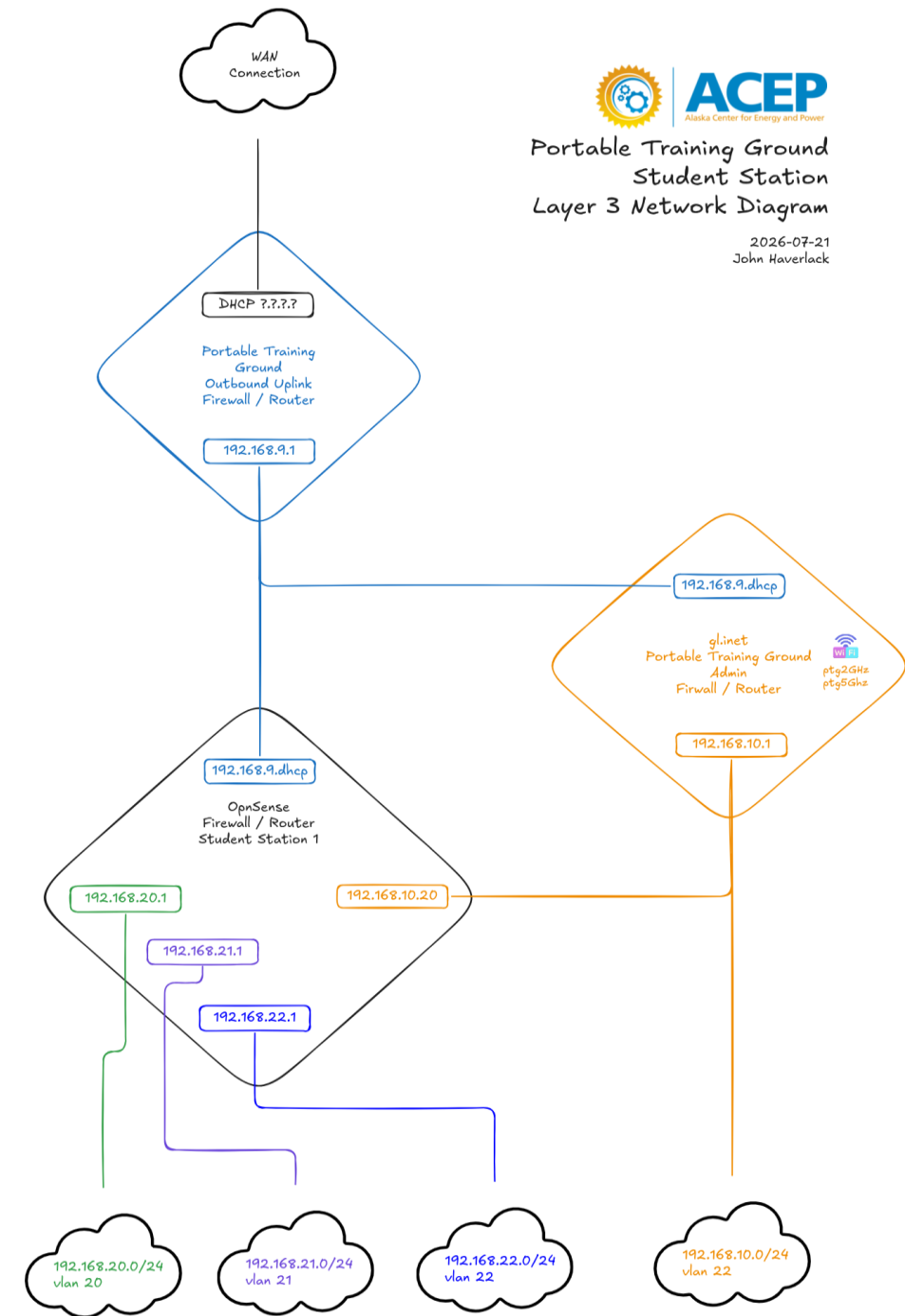
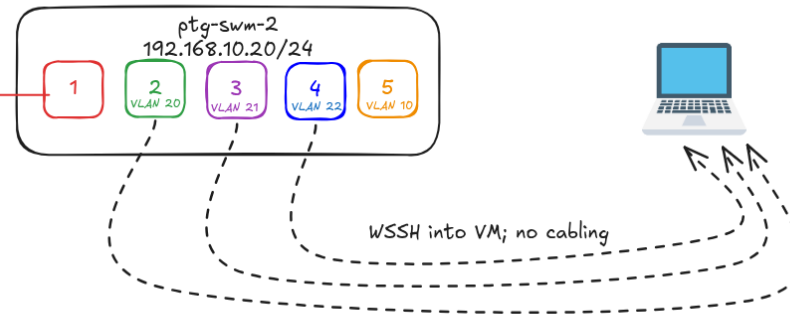
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Diagram of Our Networks

What parts can you match on these two diagrams?

- The VLAN IPv4 will be on this subnet:
-> 192.168.XX.ZZZ Where XX is your VLAN and ZZZ is the DHCP address given
- Each VLAN is one student in the WSSH

Physically Cabled to Trunk switch that carries all VLANs



Ports/Services



IP is the post office, port is the specific mailbox



Hosts specific computer program



Common ports; 443, 80, 22



Socket connection: Both IP and port connected from sender/receiver



20 Minute Break



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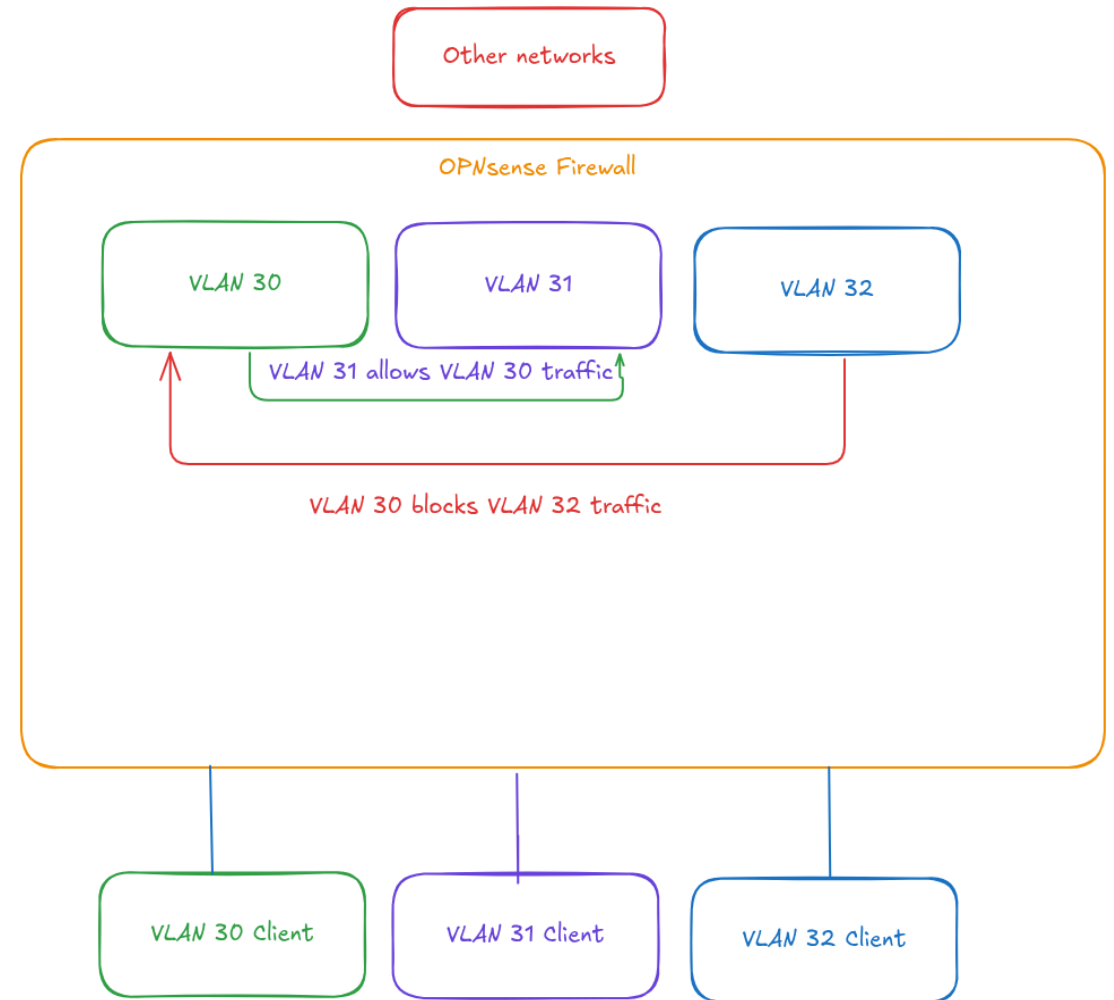
Module 2 Agenda: Firewalls

M2: *these are filler times and need to be changed per presentation, however this is intended to be a ~3 hour module.*

- **3:30pm – 4:10pm** - *What is a Firewall? Diagram Current Network, Get Into OPNsense WUI*
- **4:10pm – 4:20pm** - ^{TBD TBD} *10 minute break*
- **4:20pm – 5:20pm** - *Configure VM DHCP, DNS, and NTP, Testing With Chrony and Pings*
- **5:20pm – 6:20pm** - *Firewall Rules, Configuration, Testing*
- **6:20pm – 6:30pm** - *Feedback Form, END*

What Is A Firewall

- The bouncer at a party's front door
- Boundary of two networks, LANs or WANs in any combo
- Rules guide what's allowed; ports, subnets, etc.
- Without one, devices on your network are exposed to anything on the internet trying to reach it



Showing Rule Changes

Editing my completed OPNsense VMs



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Logging In

Your own OPNsense firewall VMs



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OPNsense VM IP	Your VM ID
	120
	121
	122
TBD	130
TBD	131
TBD	132
TBD	140
TBD	141
TBD	142
TBD	150
TBD	151
TBD	152

Everyone has this info

User: root

Password: opnsense

Enter the IP into your browser to access web interface

Your OPNsense VM will manage only your VLAN, and thus only your VM. Start up your OPNsense VM and open its console.

```
ADMIN (vtnet1) -> v4/DHCP4: 192.168.10.145/24
VLAN20 (vtnet2) -> v4: 192.168.20.1/24
WAN (vtnet0) -> v4/DHCP4: 192.168.9.207/24
```

```
HTTPS: SHA256 4A FF 05 59 D1 EC 3B CF 93 57 71 43 24 4D 82 BB
            8F FE AF AA 66 EE C5 74 67 49 27 69 6A A7 D0 FE
```

- | | |
|------------------------------|-------------------------|
| 0) Logout | 7) Ping host |
| 1) Assign interfaces | 8) Shell |
| 2) Set interface IP address | 9) pfTop |
| 3) Reset the root password | 10) Firewall log |
| 4) Reset to factory defaults | 11) Reload all services |
| 5) Power off system | 12) Update from console |
| 6) Reboot system | 13) Restore a backup |

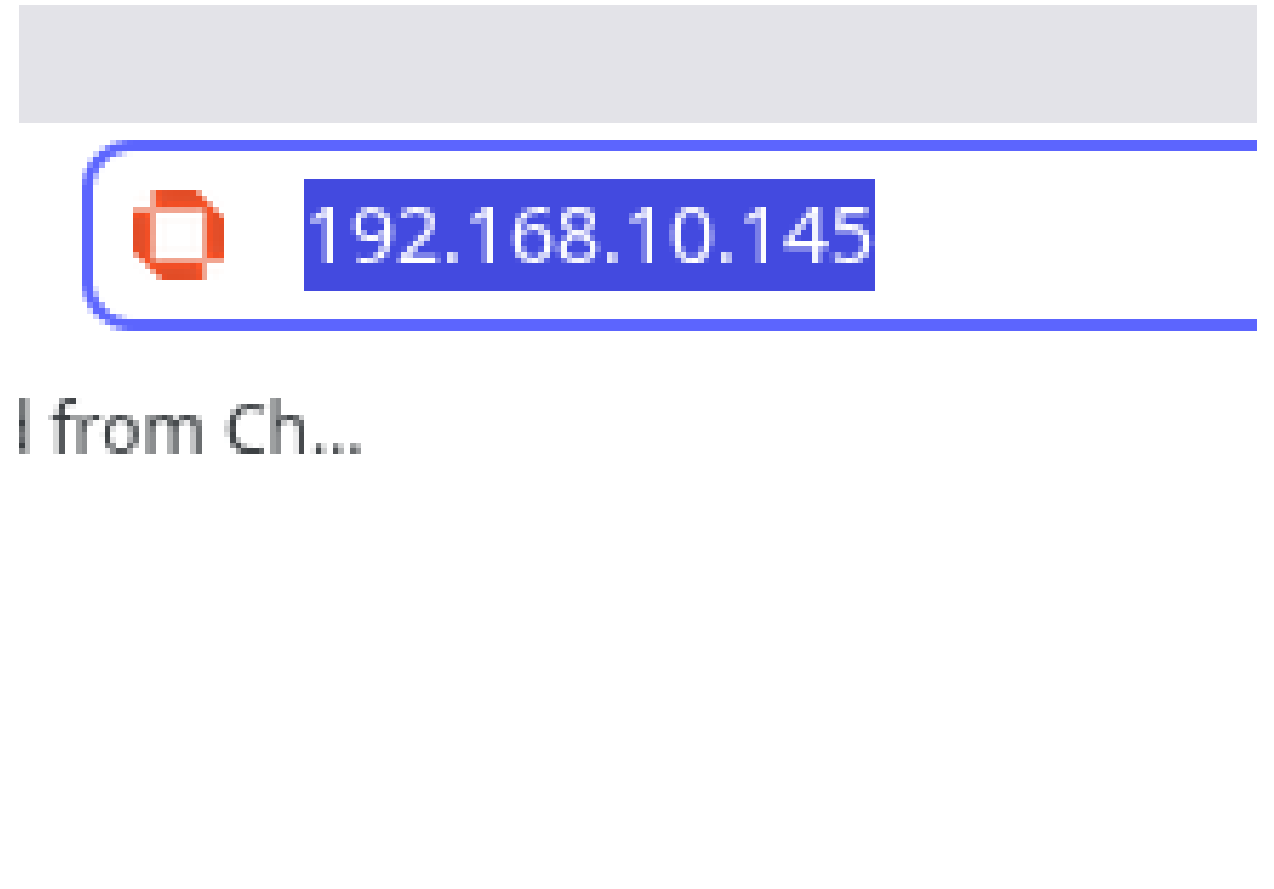
Enter an option:

- It should look like this. ADMIN is LAN on .10 subnet, WAN is .9, and your VLAN is configured to a gateway IP
- We need the ADMIN IP since that is the IP we type into the search bar to see the web interface



Access the Web User Interface

- This will be different for everyone's **OPNsense VM's!!** Insert your own
- Type **your vtnet1 LAN IP** into the search bar (not the one in the picture)
- User: root
- Password: opnsense



OPNsense®
Securing networks made easy

<

root@OPNsense.internal



Lobby

Dashboard

License

Password

Logout

Reporting

System

Interfaces

Firewall

VPN

Services

Power

Help

Starting initial configuration!



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Securing networks made easy

Welcome to OPNsense!

One moment while we start the initial setup wizard.

To bypass the wizard, click on the logo in the upper left corner.

10 Minute Break

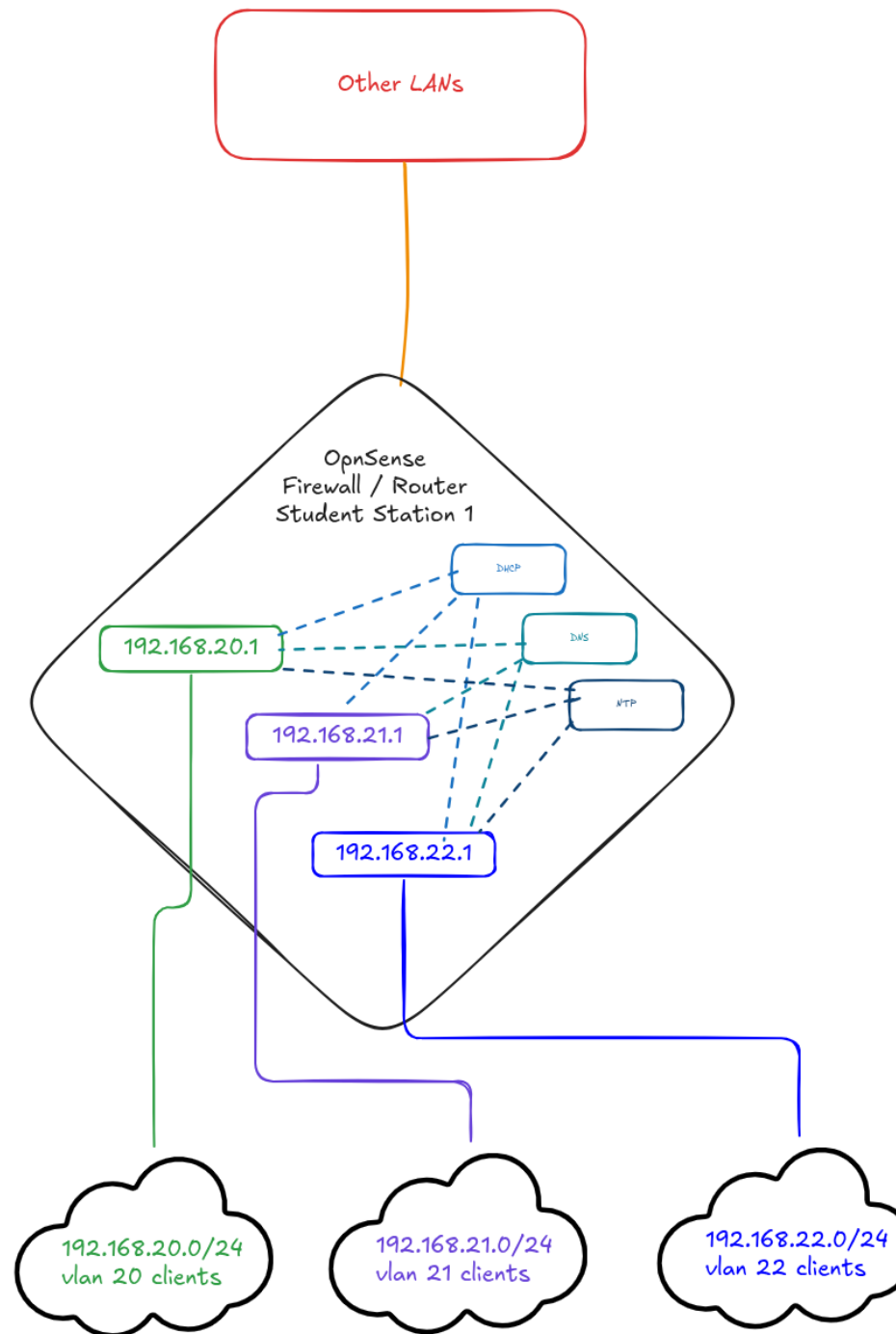


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Configuring Services

**This is what our
OPNsense firewall will
be distributing service-
wise to your host(s)**



DHCP/DNS Config



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Example

- Type `cat/etc/resolv.conf`
- Internal is a **local domain name**
- Common placeholder for devices inside a private network
- We will change it for organizational purposes

```
student4@ptg-swm5-50:~$ cat /etc/resolv.conf
# Generated by dhcpd from ens18.dhcp
# /etc/resolv.conf.head can replace this line
domain internal
nameserver 192.168.50.1
# /etc/resolv.conf.tail can replace this line
```

DHCP

In Module 1, your VM asked a DHCP server for an address.

Now, OPNsense is that server for your VLAN.

You define DHCP ranges

Your LAN clients get these IPs

What range makes a good DHCP pool?

New Device

Firewall DHCP Service

1) Discover

2) Offer

3) Request

4) Acknowledge

Device is now configured with an IP



Services: Dnsmasq DNS & DHCP

General Domains Hosts DHCP ranges DHCP options DHCP boot DHCP tags

advanced mode

▼ Default

• Enable ☒

• Interface Clear All Select All

▼ DNS

• Listen port

• DNSSEC ☐

• No hosts lookup ☐

▼ DNS Query Forwarding

• Query DNS servers sequentially ☐

• Require domain ☐

• Do not forward to system defined DNS servers ☐

• Do not forward private reverse lookups ☐

▼ DHCP

• DHCP FQDN ☒

• DHCP default domain



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Setting DHCP/DNS

- Go to Services > Dnsmasq > general
- **What should we change?**

Services: Dnsmasq DNS & DHCP

General Domains Hosts DHCP ranges DHCP options DHCP boot DHCP tags

advanced mode

▼ Default

1 Enable ☒

1 Interface ADMIN Clear All Select All

▼ DNS

1 Listen port 53053

1 DNSSEC ☐

1 No hosts lookup ☐

▼ DNS Query Forwarding

1 Query DNS servers sequentially ☐

1 Require domain ☐

1 Do not forward to system defined DNS servers ☐

1 Do not forward private reverse lookups ☐

▼ DHCP

1 DHCP FQDN ☒

1 DHCP default domain internal



Services: Dnsmasq DNS & DHCP

General Domains Hosts DHCP ranges DHCP options DHCP boot DHCP tags

advanced mode

▼ Default

1 Enable ☒

1 Interface VLAN20, VLAN21, VLAN22 Clear All Select All

▼ DNS

1 Listen port 53053

1 DNSSEC ☐

1 No hosts lookup ☐

▼ DNS Query Forwarding

1 Query DNS servers sequentially ☐

1 Require domain ☐

1 Do not forward to system defined DNS servers ☐

1 Do not forward private reverse lookups ☐

▼ DHCP

1 DHCP FQDN ☒

1 DHCP default domain ptg.acep.uaf.edu

1 DHCP local domain ☒

1 DHCP authoritative ☐

1 DHCP reply delay

OPNsense (c) 2014-2026 Deciso B.V.



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Setting DHCP/DNS

- Change the ADMIN interface to your VLAN gateway
- Helps organize devices under a common network domain.
- Makes it easier to resolve hostnames without typing the full domain.
- Does **not** change IP addresses, routing, or Internet connectivity.

- Lobby
- Reporting
- System
 - Access
 - Configuration
 - Firmware
 - Gateways
 - High Availability
 - Routes
 - Settings
 - Administration
 - Cron
 - General
 - Logging
 - Miscellaneous
 - Tunables
 - Snapshots
 - Trust
 - Log Files
 - Diagnostics
- Interfaces
- Firewall
- VPN
- Services
- Power
- Help

System: Settings: General

System full help

Hostname

ptg-OPNsense-20

Domain

ptg.acep.uaf.edu

Time zone

Etc/UTC

Language

English

Theme

opnsense

Picture

Choose File

No file chosen

Networking

Prefer IPv4 over IPv6

☒ Prefer to use IPv4 even if IPv6 is available

DNS servers

DNS Server	Use gateway
	none
	none
	none
	none
	none
	none

Edit DHCP range ×

☒ advanced mode full help ☐

i Interface	Any
i Tag [set]	None
i Start address	
i End address	
i Subnet mask	automatic
i Constructor	None
i Lease time	86400
i Domain	
i Disable HA sync	☐
i Description	



Client Ranges

- Services > Dnsmasq > ranges
- Click the + in the bottom right
- **What should we change?**



Caden's ACEP Intern Presentation

UA - Calendar - Week of July 19, 2024

tg-pve-0 - Proxmox Virtual Environment

Training Module Slides.pptx

Dnsmasq DNS & DHCP | Services

Notepad | Write your notes online

< > ↺

🔒 Not secure https://192.168.10.191/ui/dnsmasq/settings#dhcpranges

▶ 🛡️ ⚠️

📄 📁 📌 ⚙️ ☰

📦 Debian.org 📰 Latest News 📖 Help 📁 Imported from Ch...

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<

root@ptg-OPNsense-20.ptg.acep.uaf.edu 🔍

Lobby

Reporting

System

Interfaces

Firewall

VPN

Services

Captive Portal

DHCRelay

Dnsmasq DNS & DHCP

General

Domains

Hosts

DHCP ranges

DHCP options

DHCP boot

DHCP tags

Leases

Log File

Intrusion Detection

Kea DHCP

Monit

Network Time

OpenDNS

Router Advertisements

Unbound DNS

Power

Help

🔍

🎯

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🕒

🔍

🔍

🛡️

🔍

🔍

Services: Dnsmasq DNS & DHCP

▶ 🔄 ⬛

General Domains Hosts DHCP ranges DHCP options DHCP boot DHCP tags

🔍 Search

🔼 Tags & Interfaces ▾

🔄 50 ▾

☰

📄

☐	Interface	Start address	End address	Domain	Description	Commands
▼	VLAN20 ⓘ					
☐	VLAN20	192.168.20.100	192.168.20.200		DHCP for VLAN 20	✎ 📄 🗑️
▶	VLAN21 ⓘ					
▶	VLAN22 ⓘ					

« < 1 > »

Showing 1 to 3 of 3 entries + 🗑️

Apply

OPNsense (c) 2014-2026 Deciso B.V.

Lesson 2

Edit DHCP range

×

advanced mode

full help

Interface

Any

Tag [set]

None

Start address

End address

Subnet mask

automatic

Constructor

None

Lease time

86400

Domain

Disable HA sync

☐

Description

Cancel

Save



Edit DHCP range

×

advanced mode

Interface

VLAN20

Tag [set]

None

Start address

192.168.20.100

End address

192.168.20.200

Subnet mask

255.255.255.0

Constructor

None

Lease time

86400

Domain

Disable HA sync

☐

Description

DHCP for VLAN 20

Client Ranges

- Services > Dnsmasq > ranges
- Click the + in the bottom right
- **What should we change?**



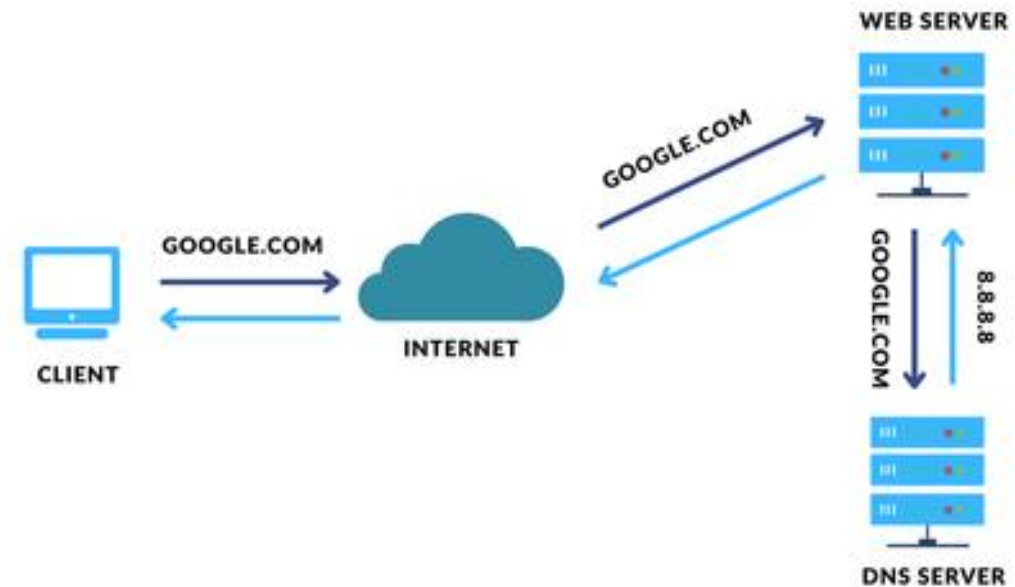
Return to Example

- Type `cat/etc/resolv.conf`
- What is the difference?
- We want meaningful naming across our network

```
student4@ptg-swm5-50:~$ cat /etc/resolv.conf
# Generated by dhcpd from ens18.dhcp
# /etc/resolv.conf.head can replace this line
domain ptg.acep.uaf.edu
nameserver 192.168.50.1
# /etc/resolv.conf.tail can replace this line
```

DNS

- OPNsense can forward requests upstream or resolve them itself
- Unbound DNS provides local DNS for LANs
- For looking up non-LAN services, OPNsense will upstream them



NTP Config



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Setting NTP

Services: Dnsmasq DNS & DHCP

General Domains Hosts DHCP ranges DHCP options DHCP boot DHCP tags

Search Tags & Interfaces 50

Interface	Type	Option	Option6	Value	Description	Commands
VLAN20	Set	ntp-server [42]		192.168.20.1	NTP Server For VLAN20	

Showing 1 to 3 of 3 entries

- Go to Services > Dnsmasq > DHCP options. Click the + in the bottom right
- Devices will synchronize to the firewall then internet
- DHCP option 42 advertises an NTP server to clients on VLAN20.
- When devices on your VLAN obtain an IP -> gateway address as time server.
- Hosts have synced clocks automatically

Setting NTP

- Use your own VLANs
- The value means we are using the gateway for devices to sync time through

Edit DHCP option ×

full help 

 Interface	VLAN20
 Type	Set
 Option	ntp-server [42]
 Option6	None
 Tag	Nothing selected
	 Clear All  Select All
 Value	192.168.20.1
 Force	☐
 Description	NTP Server For VLAN20

Cancel Save

Setting NTP

- Go to Services > Network Time > General
- The only thing we are changing on this page is to add our ntp path
- Click the + in the bottom left, and add the "ntp.alaska.edu" into the new box.
- Check "prefer" here

Services: Network Time: General

NTP Server Configuration

 Time servers

	Network	Prefer	Pool	Iburst	Do not use
	0.opnsense.pool.ntp.org	☐	☒	☐	☐
	1.opnsense.pool.ntp.org	☐	☒	☐	☐
	2.opnsense.pool.ntp.org	☐	☒	☐	☐
	3.opnsense.pool.ntp.org	☐	☒	☐	☐
	ntp.alaska.edu	☒	☐	☐	☐
					

Firewall Rules Config

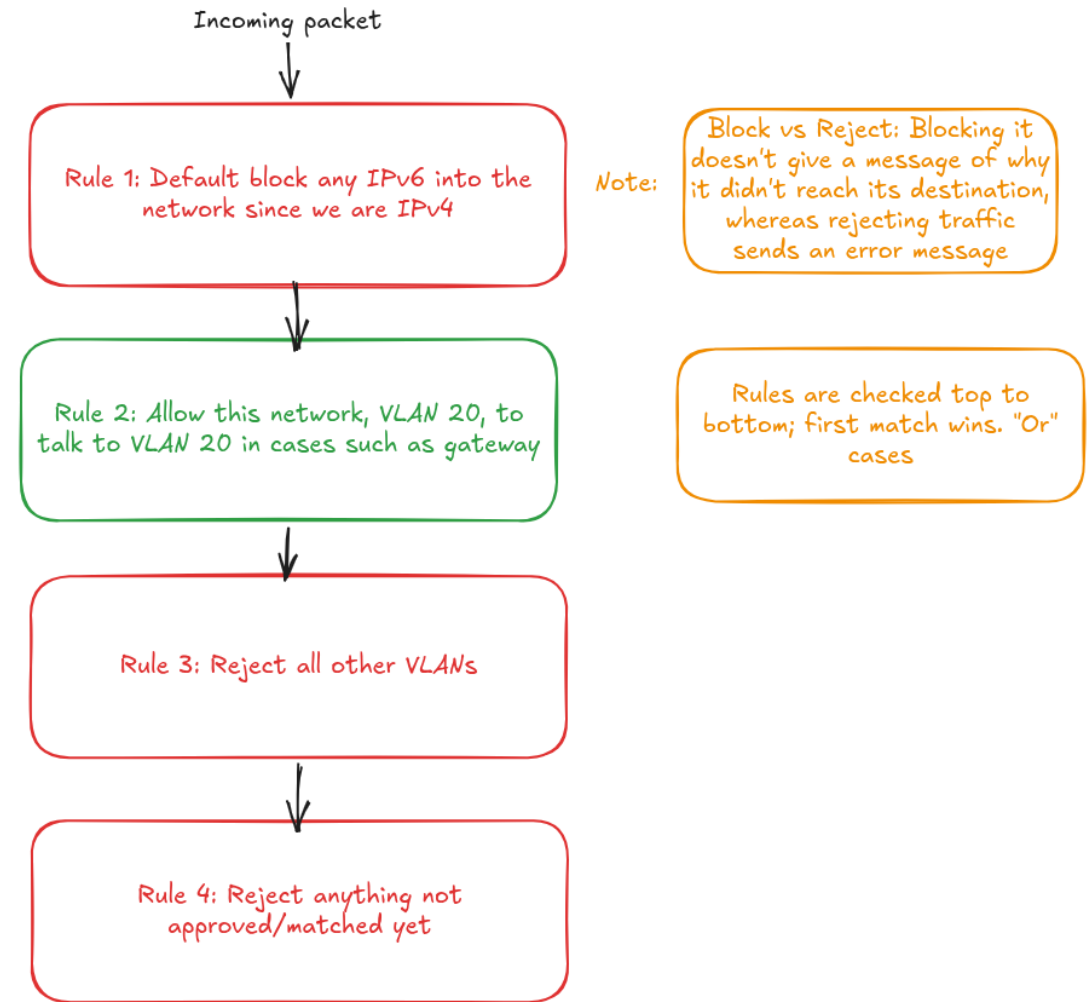


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Firewall Rules

- A firewall rule is just a statement: "For this type of traffic, from here, to there, on this port, allow or block it"
- "Or" analogy: First match wins
- Final "block all else" rule
- Good rules are specific.
- Test your rules immediately



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Practice – LAN Interface:

#	Action	Interface	Source	Destination	Port
1	Pass	LAN	LAN net	any	443
2	Pass	LAN	LAN net	any	53
3	Block	LAN	LAN net	any	80
4	Block	LAN	any	any	any (default rule)

- **Q1** A laptop on your LAN visits a website over plain HTTP (port 80). Pass or block? Which rule decides it, and why doesn't rule 1 apply, even though it's also about web traffic?
- **Q2** That same laptop launches an online game using port 25565. Pass or block? Which rule decides it?

Answers

- **Q1 is Blocked, by rule 3.**

Rule 1 only matches port 443. Port 80 is a different port, so rule 1 never applies, so every field (source, destination, port) has to match.

- **Q2 is Blocked, by the bottom "catch all" rule.**

Nothing in rules 1 through 3 mentions port 25565, so nothing matches until the catch-all at the bottom. No matching allow rule means it is blocked.

Practice:

#	Action	Interface	Source	Destination	Port
1	Pass	WAN	192.168.20.50	LAN net	22
2	Block	WAN	any	LAN net	22
3	Pass	WAN	any	LAN net	443
4	Block	WAN	any	any	any (<i>default rule</i>)

- Q1** A connection arrives from 192.168.20.50 trying SSH (port 22). Then a connection arrives from 192.168.50.9, also trying SSH. What happens to each, and why are the outcomes different?
- Q2** Suppose rules 1 and 2 were swapped, "block SSH from anyone" listed first, "allow SSH from 192.168.20.50" listed second. Would the trusted admin at 192.168.20.50 still get in?

Answers

- **Q1 192.168.20.50 passes** (matches rule 1 exactly because it's the right source and right port). **192.168.50.9 is blocked** (fails rule 1's source check, falls through to rule 2's blanket block). Source is just as much a required match as port.
- **Q2 No, as the admin gets blocked too.**
"Block SSH from anyone" matches anyone, including 192.168.20.50. The first match wins, so OPNsense never reaches the specific allow rule below it. Order alone can defeat an otherwise-correct rule.

Setting Up Some Baseline Rules

- Setting up a baseline for you to edit afterwards that will give you familiarity
- Navigate to: Firewall > Rules [new]
- Click the "all rules" bar on top, then select "VLANXX"
- **One person in your group should do this for each VLAN**

Firewall: Rules [new]

2 All rules

Categories

Inspect Tree

Search

50

Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands
IPv4	*	ADMIN network	*	*	*	*	Default allow LAN to any rule	   
IPv6	*	ADMIN network	*	*	*	*	Default allow LAN IPv6 to any rule	   

Showing 1 to 2 of 2 entries

+     

It Will Be Blank

Firewall: Rules [new]

VLAN50

Categories

Inspect

Tree

Q

Search

↺

50

≡

↻

☐		☒	Icons		Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands
No results found													

«

<

1

>

»

Showing 0 to 0 of 0 entries

+

☑

□

🗑

⬆

📄

Apply

First Rule



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Edit rule

advanced mode full help

Organisation

Enabled ☒

Categories Nothing selected
Clear All Select All

Description

Interface

Invert Interface ☐

Interface VLAN50
Clear All Select All

Filter

Quick ☒

Action Pass

Direction In

Version IPv4

Protocol any

Invert Source ☐

Source any

Cancel

Save

- We want to deny all IPv6 traffic on this VLAN since we won't be using it
- **What should we change?**

First Rule



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Edit rule

advanced mode

full help

Organisation

Enabled

☒

Categories

Nothing selected

Clear All

Select All

Description

Interface

Invert Interface

☐

Interface

VLAN50

Clear All

Select All

Filter

Quick

☒

Action

Pass

Direction

In

Version

IPv4

Protocol

any

Invert Source

☐

Source

any

Cancel

Save



Edit rule

advanced mode

full help

Organisation

Enabled

☒

Categories

Nothing selected

Clear All

Select All

Description

Default deny LAN IPv6 to any ...

Interface

Invert Interface

☐

Interface

VLAN50

Clear All

Select All

Filter

Quick

☒

Action

Reject

Direction

In

Version

IPv6

Protocol

any

Invert Source

☐

Source

VLAN50 network

Cancel

Save

Second Rule



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Edit rule

Categories

Nothing selected

Clear All

Select All

Description

Interface

Invert Interface

☐

Interface

VLAN50

Clear All

Select All

Filter

Quick

☒

Action

Pass

Direction

In

Version

IPv4

Protocol

any

Invert Source

☐

Source

any

Source Port

any

Invert Destination

☐

Destination

any

Cancel

Save

- We want to allow VLAN50 to talk to itself in all cases (e.g. gateway); not usually needed as a command
- **What should we change?**

Lesson 3

Second Rule



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Edit rule

Categories Nothing selected
✖ Clear All ✔ Select All

Description

▼ **Interface**

Invert Interface ☐

Interface VLAN50
✖ Clear All ✔ Select All

▼ **Filter**

Quick ☒

Action Pass

Direction In

Version IPv4

Protocol any

Invert Source ☐

Source any

Source Port any

Invert Destination ☐

Destination any

Cancel

Save



Edit rule

Categories Nothing selected
✖ Clear All ✔ Select All

Description Allow VLAN 50 to talk to VLAN...

▼ **Interface**

Invert Interface ☐

Interface VLAN50
✖ Clear All ✔ Select All

▼ **Filter**

Quick ☒

Action Pass

Direction In

Version IPv4

Protocol any

Invert Source ☐

Source VLAN50

Source Port any

Invert Destination ☐

Destination This Firewall

Cancel

Save

Third Rule



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Edit rule

Categories

Nothing selected

Clear All

Select All

Description

Interface

Invert Interface

☐

Interface

VLAN50

Clear All

Select All

Filter

Quick

☒

Action

Pass

Direction

In

Version

IPv4

Protocol

any

Invert Source

☐

Source

any

Source Port

any

Invert Destination

☐

Destination

any

Cancel

Save

We want to deny traffic from all other VLANs?

What should we change?

Third Rule



ACEP
Alaska Center for Energy and Power

Edit rule ✕

Categories Nothing selected
✕ Clear All ✓ Select All

Description

Interface

Invert Interface ☐

Interface VLAN50
✕ Clear All ✓ Select All

Filter

Quick ☒

Action Pass

Direction In

Version IPv4

Protocol any

Invert Source ☐

Source any

Source Port any

Invert Destination ☐

Destination any

Cancel Save



Edit rule ✕

Description Reject all other VLANs

Interface

Invert Interface ☐

Interface VLAN50
✕ Clear All ✓ Select All

Filter

Quick ☒

Action Reject

Direction In

Version IPv4

Protocol any

Invert Source ☐

Source TG_VLANs

Source Port any

Invert Destination ☐

Destination VLAN50

Destination Port any

Cancel Save

Fourth Rule



ACEP
Alaska Center for Energy and Power

Edit rule ✕

Categories

Nothing selected

✕ Clear All ✔ Select All

Description

▼ Interface

Invert Interface

☐

Interface

VLAN50

✕ Clear All ✔ Select All

▼ Filter

Quick

☒

Action

Pass

Direction

In

Version

IPv4

Protocol

any

Invert Source

☐

Source

any

Source Port

any

Invert Destination

☐

Destination

any

Cancel

Save

- We want to deny all other traffic not specifically approved
- **What should we change?**

Fourth Rule



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Edit rule

Categories

Nothing selected

Clear All

Select All

Description

Interface

Invert Interface

☐

Interface

VLAN50

Clear All

Select All

Filter

Quick

☒

Action

Pass

Direction

In

Version

IPv4

Protocol

any

Invert Source

☐

Source

any

Source Port

any

Invert Destination

☐

Destination

any

Cancel

Save



Edit rule

Description

Reject anything not approved...

Interface

Invert Interface

☐

Interface

VLAN50

Clear All

Select All

Filter

Quick

☒

Action

Reject

Direction

Both

Version

IPv4

Protocol

any

Invert Source

☐

Source

any

Source Port

any

Invert Destination

☐

Destination

any

Destination Port

any

Cancel

Save

Our Baseline For All 3 VLANs Is Set



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Testing!

We'll log into the VMS now that firewall services are configured



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192.168.10.10:8006

Type this into your
search bar



Ask anything, find anything...



tg-pve-0 - Proxmox Virtual En

192.168.10.10:8006/#v1:0:18:4:.....

DocumentationCreate VMCreate CT

Server ViewDatacenter

Proxmox VE Login

User name: student1

Password: student1

Realm: Linux PAM standard authentication

Language: English - English

Save User name: ☐ Login

TasksCluster log

Start Time ↓	End Time	Node	User name	Description	Status
--------------	----------	------	-----------	-------------	--------

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Lesson 1

ENBluetoothWiFiSpeakerBattery11:15

Everyone's username and password will be handed out and be student# where # is assigned to you

tg-pve-0 - Proxmox Virtual Envi

Not secure https://192.168.10.10:8006/#v1:0:=qemu%2F101:4:5:::29::

Debian.orgLatest NewsHelpImported from Ch...

PROXMOX

Virtual Environment 9.2.3

Search

DocumentationCreate VMCreate CTstudent1@pam

Server View

Datcenter

tg-pve-0

101 (ptg-V20-M1)

102 (ptg-V20-M1)

105 (ptg-V20-M1)

106 (ptg-V20-M1)

107 (ptg-V20-M1)

108 (ptg-V20-M1)

109 (ptg-V20-M1)

110 (ptg-V20-M1)

111 (ptg-V20-M1)

112 (ptg-V20-M1)

113 (ptg-V20-M1)

114 (ptg-V20-M1)

120 (ptg-swm2-20)

121 (ptg-swm2-21)

122 (ptg-swm2-22)

998 (ptg-OPNsense)

999 (ptg-dhcp)

100 (ptg-V20-M1)

103 (deb13-tmpl-260610)

tg-pve-1

tg-pve-2

tg-pve-3

Virtual Machine 101 (ptg-V20-M1) on node 'tg-pve-0'

Summary

Console

Hardware

Cloud-Init

Options

Task History

Backup

Replication

Snapshots

Firewall

Take SnapshotRollbackViewRemove

The current guest configuration does not support taking new snapshots

Name	RAM	Date/Status	Description
pre-static	Yes	2026-07-07 09:53:03	
post-static	Yes	2026-07-07 09:58:46	
NOW			You are here!

StartShutdownMigrateConsoleMoreHelp

TasksCluster log

Start Time	End Time	Node	User name	Description	Status
------------	----------	------	-----------	-------------	--------

Click on "tg-pve-0" on the left side once. Then, you should see this.

Student	VM ID
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx
StudentName	192.168.xx.xxx

We will pass out sheets with your username, password, and VM for this part.

Note: all challenges only need you to edit the VLAN interfaces for your OPNsense. They will only require you to work in your current groups

Challenge 1:

Goal: In your groups, get **Teammate A** able to ping **Teammate B**. **Teammate B** should still *not* be able to ping **Teammate A**.

Note: all challenges only need you to edit the VLAN interfaces for your OPNsense. They will only require you to work in your current groups

Challenge 2:

Goal: Make **Teammate B** and **Teammate C** able to ping each other, both directions. **Teammate A** should be unreachable by, and unable to reach, either of them.

Note: all challenges only need you to edit the VLAN interfaces for your OPNsense. They will only require you to work in your current groups

Challenge 3:

Goal: A rule on **Teammate A's** tab is currently too generous, as it lets A reach **both** B and C. Create the "generous rule", then change it so only B is reachable from A.

Feedback Form



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